CNPI

METHODS PAPER · VERSION 1.2

Corporate Nature Pressure Index

How we compare India's 500 largest listed companies on the environmental pressure they report, and what that pressure would cost society

Urvara.Life with Landstack · for ILDC 2026, Bhubaneswar

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Status of this document. This paper sets out the method behind the Corporate Nature Pressure Index (CNPI), version 1.2. The index used to be called the Corporate Land and Biodiversity Index (CLBI), and the output files still carry the `CLBI_v1_` prefix. We computed every number in this paper from the build of 2026-10-28 12:28:30 UTC (`outputs_v12/build_stats.json`), made after an independent review of data and method (Appendix F), or we quote it from a cited source. Appendix E shows how we computed the less obvious numbers. The paper adds no estimate of a company's environmental performance to what the company itself reported.

Two products, kept apart. The paper describes two products. (1) The **CNPI physical pressure index** ranks reported physical quantities within peer groups. (2) The **illustrative nature cost** multiplies the same quantities by published damage prices and expresses the result as a share of EBITDA. The cost never enters a rank, and a rank never uses a price.

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The method in brief

This chapter gives the whole method in a few pages: the starting idea and the reason for the change of name, the formula, the sources, the adjustments we make to reported figures, the rules that keep companies out of a rank, the path from 500 companies to 409, and the rules that apply to particular sectors. The numbered sections give the detail and the evidence.

From land and biodiversity to nature pressure

We set out to build a **Corporate Land and Biodiversity Index** (CLBI). The aim was to compare India's largest listed companies on the land they hold, where their sites sit against sensitive habitats, and what their operations do to biodiversity. Land-use change is the largest direct driver of biodiversity loss on land (IPBES 2019), so land was the natural place to start.

We could not find the data to do it.

- The BRSR, the sustainability report SEBI requires of large listed companies, has no field for hectares, coordinates, land-use change or restoration. It asks for the *number* of plants and offices and for a free-text table of operations in or near ecologically sensitive areas (Section 10.1).
- An independent search of annual reports, mining-lease tables and environmental-clearance letters for all 500 companies found a company-wide figure for land held for **13 companies (2.6%)**, after we checked each figure by hand. It found printed coordinates for **10 of 326** named sites.
- Of 87 statements about operating near a sensitive area, 18 from 14 companies say yes. None gives coordinates.

A land or biodiversity score built on 13 companies would rank disclosure, and only for a handful of firms. In version 1.1, restoration, recycling and renewable electricity earned up to 10 bonus points on top of the pressure score. With so little land evidence to stand on, and with recycling and renewables now netted inside their own pressures, we set that bonus to zero in version 1.2.

Almost every company does report four quantities in a machine-readable filing: greenhouse-gas emissions, energy from non-renewable sources, water consumed, and waste it did not recover. IPBES (2019) lists climate change, pollution and the use of natural resources among the direct drivers of nature loss, and these four quantities measure them in a company's own operations. We built the index on them and renamed it the **Corporate Nature Pressure Index** (CNPI). The name states the scope: pressure the company itself reports. CNPI does not measure the state of nature or a company's biodiversity impact. Land stays in the index as evidence beside the rank (Section 10), with a site-level biodiversity metric ready for the day companies report site areas and coordinates (Section 10.3).

The formula

For company i in peer group g and pressure p:

1. **Net pressure** $Q_{i,p}$, from the company's FY2025-26 BRSR (table below).
2. **Intensity**, per ₹ crore of revenue from operations on the same boundary as the BRSR:

$$I_{i,p} = \frac{Q_{i,p}}{\mathrm{Rev}_i}$$

3. **Score** within the group. Sort the k companies with a usable intensity from lowest to highest; $\rho_{i,p}$ is the company's position, and tied values share the average position:

$$s_{i,p} = 100 \times \frac{k - \rho_{i,p}}{k - 1}$$

   The lowest intensity scores 100 and the highest 0.
4. **Final score**, the unweighted mean over the pressures the group requires (four, or three where pollution is unscored):

$$F_i = \frac{1}{|\mathcal{R}_g|} \sum_{p \in \mathcal{R}_g} s_{i,p}$$

5. **Rank** within the group by F_i rounded to six decimals. Rank 1 is the lowest reported pressure per rupee of revenue.

Pressure	BRSR item	Gross figure	Offset that reduces it	Net figure we score
Climate	P6 EI 7	Scope 1 + Scope 2, tCO2e	none	Scope 1 + Scope 2
Energy	P6 EI 1	total energy, GJ	energy from renewable sources	non-renewable energy
Water	P6 EI 3	withdrawal, kL	water returned	consumption, capped at withdrawal
Pollution	P6 EI 9	waste generated, t	waste recovered	generated minus recovered, never below 0
Air (separate rank)	P6 EI 6	NOx + SOx + PM, t	none	NOx + SOx + PM

An offset reduces only the pressure it belongs to. Renewable power cannot make up for unrecovered waste. Sections 4 and 7 give the full definitions.

Where the data come from

Source	What we take from it	Coverage
BRSR XBRL file filed with NSE	the four pressures and air, with the unit the filing declares	465 of 500 companies have a parseable file; 84.2% of the 4,268 core values are machine-readable with an explicit unit (tier 1)
Filed BRSR PDF	unit checks on XBRL values (tier 2, 8.2%); corrections of unit slips (tier 5, 1.0%)	Climate needs the most checking: 246 filings tag Scope 1 and 2 with the ambiguous unit “MtCO2e”
Annual and sustainability reports	values the XBRL left blank, each with a URL and a verbatim quote (tier 3, 5.6%)	1,007 of 3,870 candidate values accepted
Annual financial filing with NSE	revenue from operations on the BRSR’s own boundary	usable for 487 companies
NSE Indices classification and BRSR product lines (NIC codes, turnover shares)	peer groups	29 groups; 54 companies moved to the group of their main activity
Annual reports, mining-lease tables, environmental-clearance letters	land and site evidence, outside the score	all 500 searched
Published damage prices	the illustrative nature cost, outside the score	Section 11

Every value carries its source URL, the XBRL fact or the PDF page with a verbatim quote, the transformation applied and the SHA-256 fingerprint of the source file (`CLBI_v1_company_data.csv`). Sections 3.1 to 3.7 describe each route.

Adjustments to reported figures

We change a reported figure only by a stated rule, and the dataset shows the figure before and after.

- **Offsets.** Renewable energy, water returned and waste recovered come off their own pressure, in full (Section 4.6).
- **Water above withdrawal.** Where consumption is 1.05 to 10 times withdrawal, the company has counted recycled water as consumed. We score withdrawal (15 companies).
- **Water at exactly 20% of withdrawal.** The company has applied an assumed 80% discharge in place of a measurement. We give no offset and score withdrawal (30 companies, 24 ranked).
- **Waste recovered above generated**, up to 1.5 times: the net pressure is zero.
- **Zero Scope 2** stands where renewables supply at least 90% of energy or electricity (7 of 11 zero values).

- **Energy tables** corrected to the filed PDF count as a set: if the corrected total fails, we drop the corrected split too (13 companies corrected).
- **Seawater**, for the nature cost only. We remove the seawater inside reported consumption from priced water for 7 large water users. Ranks are unaffected.

Values we hold back, and companies we caution

We hold back a value that looks wrong, even if the filing prints it, until the company confirms it. A company with a held-back value on a pressure its group scores loses its rank. Section 6 sets out each rule.

Rule	Test	Effect in this build
V1 Reported zero	zero gross figure, or zero Scope 1 or Scope 2 on its own	marked "to confirm"; 13 companies lose their rank
V3 Extreme value	gross intensity above 300 times, or below 1/300 of, the peer median	16 core values held back
V4 Electricity bill	Scope 2 implies a grid bill larger than revenue	2 companies
V5 Grid electricity	Scope 2 implies more than 1.5 times total energy	3 companies
V6 Water	consumption above 10 times withdrawal	4 companies
V8 Waste	recovered above 1.5 times generated	4 companies
V11 Revenue	revenue year or boundary does not match the BRSR	13 companies
V12 Scope 1 and fuel	Scope 1 above 0.12 t per GJ of fossil energy, outside process-emission groups	caution, rank kept; 15 companies
V13 Naming caution	unusual business model or boundary, or a value 100 to 300 times from the peer median	caution, rank kept; 31 ranked companies

A naming caution leaves the rank alone. It tells you not to single the company out by name, for example as a sector leader, until someone checks it.

From 500 companies to 409

Each company passes through the same tests in the same order. We count it at the first test it fails.

Test	Remaining	Dropped
Universe: top 500 NSE companies by average market capitalisation, January–June 2026 (AMFI)	500	
A FY2025-26 BRSR source covers the required pressures	475	25
Every required pressure reported	464	11
Revenue matched to the BRSR year and boundary	463	1
No held-back value on a required pressure	442	21
No reported zero awaiting confirmation	429	13
Peer group and NSE sector each have at least 8 members	416	13
At least 8 companies in the group with complete, clean data: **ranked**	**409**	7

The published exclusion list gives each of the 91 companies one main reason instead: a held-back value (24), a peer group too small to rank (25), a pressure not reported (16), revenue that does not match the BRSR (13), or a zero awaiting confirmation (13). The two views count the same 91 companies in different orders (Section 8). The 409 ranked companies sit in 24 peer groups.

Rules for particular sectors

Groups	Rule	Reason
All	Compare a company only with peers whose main business is of the same kind	Intensities differ between sectors by three to four orders of magnitude; a cross-sector rank would rank sectors

Groups	Rule	Reason
Power; non-ferrous metals and mining; construction and engineering; real estate	Pollution unscored; three pressures make the score	Companies put fly ash, overburden and tailings, or construction debris inside the waste table or leave them out (rule V9)
Cement; steel; non-ferrous metals and mining; commodity chemicals and fertilisers; oil, gas and fuels	Exempt from the Scope 1 fuel check	Process emissions are expected
Cement; power	A second view per tonne of cement and per MWh generated	A recognised physical yardstick that companies disclose (Section 9)
Five financial-services groups	Scored on their operational footprint and flagged as not comparable with the real economy; nature cost shown as a share of revenue	Financed impacts fall outside the index
Real estate	Land banks reported in their own category (R), apart from operating land	Land held for development differs from land used in operations
Building products; diversified conglomerates; investment and holding companies; textiles and apparel; telecom and media	Listed with scores where available, not ranked	Fewer than 8 members, or fewer than 8 with complete data

Two products, kept apart

The **pressure index** ranks physical quantities. The **illustrative nature cost** multiplies the same quantities by published damage prices: carbon at the share of climate damage borne in India (₹5,400 per tCO2e), water at a scarcity value (₹358 per kL), non-renewable energy at a depletion value (₹80 per GJ), unrecovered waste at ₹600 per tonne and air pollutants at Indian health-damage values. We report it as a share of EBITDA, by peer group, with two sensitivity lenses (Section 11). A price never enters a rank.

Changes from version 1.1

Area	Version 1.1	Version 1.2
Name	Corporate Land and Biodiversity Index	Corporate Nature Pressure Index
Pressures	three (climate, water, waste), gross, plus up to 10 bonus points for restoration, recycling and renewable electricity	four (climate, energy, water, pollution), net: each offset reduces its own pressure; bonus retired
Land	restoration bonus; land priced in a pilot	evidence only, no score and no price
Peer groups	NSE sectors, financial services split; at least 10 complete companies; 373 ranked in 16 groups	29 groups by main activity; at least 8; 409 ranked in 24 groups
Validation	300× rule, zero check, Scope 2 bill test	adds V2, V5, V7, V9 (pollution unscored in four groups) and V12; the NIC-based “trading” flag retired
Nature cost	carbon at the global social cost (₹22,600 per t), about 98% of the total; water ₹40 per kL; waste not priced	carbon at the India-borne share (₹5,400 per t); water at scarcity value (₹358 per kL); energy depletion and waste priced; shown as a share of EBITDA; seawater removed

1 Summary

500 companies in the universe, FY2025-26 BRSR	**409** ranked, in 24 peer groups	**91** listed with a stated reason for not being ranked	**29** reported values held back as implausible or inconsistent
0.4–316% range of peer-group medians of illustrative nature cost, as a share of EBITDA	**68.8%** of the costed total is carbon, at the India-borne damage price	**13** companies with a company-wide figure for land held	**10** of 326 named operating sites with printed coordinates

The index. CNPI compares each of India's 500 largest NSE-listed companies with peers in the same line of business. It asks one question of each company: how much pressure on nature did you report in FY2025-26 for each ₹ crore of revenue? It uses four pressures, all from the company's own Business Responsibility and Sustainability Report (BRSR):

- **climate**: Scope 1 plus Scope 2 greenhouse-gas emissions (tCO2e);
- **energy**: energy from non-renewable sources (GJ);
- **water**: water consumed (kL);
- **pollution**: waste generated minus waste recovered (t).

Renewable energy, water returned and waste recovered reduce the pressure they belong to, in full. A company cannot trade one of them against a different pressure.

Scoring. We divide each pressure by revenue. Within a peer group, the company with the lowest intensity scores 100, the highest scores 0, and the rest sit at even steps by position. We average the four scores with equal weights and rank companies within their group. Each company needs all four pressures. The exception is pollution, which we leave unscored in four groups (power; non-ferrous metals and mining; construction and engineering; real estate) because companies there report their largest waste streams in different ways.

Results. We rank **409 of the 500 companies in 24 peer groups**. We list the other 91 with the reason: a value held back as implausible or inconsistent (24 companies), a peer group too small to rank (25), a pressure not reported (16), a revenue figure that does not match the BRSR year or boundary (13), or a reported zero awaiting confirmation (13). We rank air pollution (NOx + SOx + particulate matter) on its own for 178 companies in 14 groups.

Outside the ranks.

- *Land and sites.* An independent search of all 500 companies found a company-wide figure for land held for 13 companies, after we checked each figure by hand, and printed coordinates for 10 of 326 named sites. We report land as evidence and do not score it (Section 10).
- *Illustrative nature cost.* We multiply each reported quantity by a published damage price, express the total as a share of EBITDA, and report peer-group medians with their interquartile range. The medians run from 0.37% of EBITDA in IT services and software to 316.0% in Cement. Across all 370 non-financial companies with positive EBITDA the median is 4.4% (3.3% and 4.8% under the two sensitivity lenses). Carbon, priced at the share of climate damage borne in India, makes up 68.8% of the costed total; water, priced by scarcity, 11.1%; energy depletion 9.5%; air 9.3%; waste 1.2% (Section 11). These figures never enter a rank.

Reading a rank. Rank 1 means the company reported the lowest pressure per rupee of revenue among its peers. The rank says nothing about the size of its absolute footprint, its effect on biodiversity, or wrongdoing by any other company. CNPI measures drivers of nature loss in a company's own operations. Biodiversity impact, supply chains, products in use and what a bank finances fall outside it.

Robustness. Ranks change little when we drop one pressure (median Spearman correlation 0.93 across groups; lowest 0.81, in transport and logistics) or when we use gross measures in place of net ones (median 0.93; lowest 0.79, in IT services and software). They change more when profit replaces revenue as the denominator (median 0.82; lowest 0.43, in transport and logistics).

> WHAT THIS MEANS.
> The index compares like with like on what companies themselves report, and you can trace every number to a filing and every exclusion to a stated rule. Its main finding concerns disclosure as much as performance. We cannot yet rank one company in five, and companies do not report most of the data a full nature index needs: site locations, land area, air pollution in tonnes (Section 14).

2 Scope and universe

Universe. The universe is the 500 NSE-listed companies with the highest six-month average full market capitalisation over January–June 2026, as published by the Association of Mutual Funds in India (AMFI) (`CLBI_Dated_Top500_JanJun2026_v1.csv`). The list contains no real-estate or infrastructure investment trusts.

Reporting year. FY2025-26, 1 April 2025 to 31 March 2026. The XBRL parser accepts a fact if its context covers this period and no other (Section 3.1).

Unit of analysis. We assess the listed entity on the boundary of its own BRSR. Each BRSR states whether it covers the listed company alone (standalone) or the group (consolidated). Of the 500 companies, 362 report standalone, 111 consolidated, and 27 give no boundary. Among the 409 ranked companies the split is 312, 96 and 1. We take revenue on the same basis as the company's BRSR (Section 3.6). For about three-quarters of the index, the pressures cover the listed parent alone.

What CNPI measures. CNPI measures *pressures*: reported quantities of the direct drivers of nature loss that IPBES (2019) identifies, here climate change, pollution and resource use, in a company's own operations. It does not measure the state of nature or change in it. Company disclosures cannot yet measure land-use change, the driver IPBES ranks first on land (Section 10).

Exclusions.

- Scope 3 emissions and every other value-chain pressure.
- The effects of products in use or at end of life.
- Financed and insured impacts. The financial-services peer groups show the operational footprint of banks, lenders, insurers and intermediaries (offices and data centres), and a flag (`financials_flag`) marks them as not comparable with the real economy.
- Any estimate, model output or sector average in place of a missing value. A blank stays blank.

3 Data sources and lineage

We record every value in `CLBI_v1_company_data.csv`, one row per company and field. Each row holds the value, unit, flag, confidence tier (`source_status`), source URL, the XBRL fact identifier or the PDF page with a verbatim quote (`source_field`), the transformation applied, and the SHA-256 fingerprint of the source file. All 5,949 reported values carry a source URL; a value reconciled against a PDF points to the BRSR filing.

3.1 BRSR XBRL filings

The primary source is the BRSR XBRL file each company files with NSE (SEBI 2021; SEBI 2023). Of the 500 companies, 465 had a parseable FY2025-26 BRSR XBRL file. The CLBI XBRL parser v1.0.0 read them (`normalized_companies.json`) under four rules:

- a concept qualifies if it sits in the SEBI `in-capmkt` taxonomy namespace with a local name in Table 3.1;
- the context must be the reporting entity, for 1 April 2025 to 31 March 2026, with no dimensions, segments or restatement markers;
- the parser leaves conflicting duplicate facts unresolved and does not average them;
- the parser takes a unit from the declared XBRL measure (for example GJ, TJ, MWh, kL, tCO2e) and never guesses one.

Table 3.1. BRSR XBRL concepts used. All are Principle 6 essential indicators (EI) of the BRSR format.

CNPI key	XBRL concept (`in-capmkt`)	BRSR item	Normalised unit
`scope1_tco2e`	`TotalScope1Emissions`	P6 EI 7	tCO2e
`scope2_tco2e`	`TotalScope2Emissions`	P6 EI 7	tCO2e
`total_energy_gj`	`TotalEnergyConsumedFromRenewableAndNonRenewableSources`	P6 EI 1	GJ
`renewable_energy_gj`	`TotalEnergyConsumedFromRenewableSources`	P6 EI 1	GJ
`nonrenewable_energy_gj`	`TotalEnergyConsumedFromNonRenewableSources`	P6 EI 1	GJ
`renewable_electricity_share_pct`	derived: renewable ÷ total electricity (`TotalElectricityConsumptionFrom…Sources`)	P6 EI 1	%
`water_withdrawal_m3`	`TotalVolumeOfWaterWithdrawal`	P6 EI 3	kL (= m^3)
`water_consumption_m3`	`TotalVolumeOfWaterConsumption`	P6 EI 3	kL
`waste_generated_tonnes`	`TotalWasteGenerated`	P6 EI 9	t
`waste_recovered_tonnes`	`TotalWasteRecovered`	P6 EI 9	t
`nox_tonnes`, `sox_tonnes`, `pm_tonnes`	`NOx`, `SOx`, `ParticulateMatter`	P6 EI 6	t
`scope3_tco2e` (reported, not scored)	`TotalScope3Emissions`	P6 Leadership Indicator 2	tCO2e
`revenue_inr_cr`	`RevenueFromOperations`	Section A	₹ crore

Source: parser metric dictionary (`parser/metric_dictionary.md`). `build_clbi.py` labels the three energy facts in `source_field` with the short name `TotalEnergyConsumption…`; the parser reads the concepts in this table.

3.2 Units

The parser converts declared units to the base unit with fixed multipliers (for example 1 TJ = 1,000 GJ, 1 MWh = 3.6 GJ, 1 ML = 1,000 kL). Two cases needed a rule:

- **"MtCO2e".** Many filings tag Scope 1 and 2 with "MtCO2e", which could mean metric tonnes or million tonnes, and the parser leaves these unresolved. Of the 476 reported Scope 1 values, 212 are tier-1 XBRL values with an explicit unit (Table 3.2). We resolved the rest by checking the filed PDF (tier 2), from a public report with a quote (tier 3), or with **rule R1**, which reads "MtCO2e" as metric tonnes CO2e, the unit the BRSR form prescribes (tier 4).

- **Air pollutants per year. Rule R2** reads “t/year” or “kg/year” as an annual mass in tonnes.

`inputs/unit_backfill.csv` holds 670 candidate rows for these rules (R1: 490; R2: 180). A value read from the filed PDF with a verbatim quote takes precedence over a rule. In the final build the rules supply 51 values: 41 Scope 1 or Scope 2 values and 10 air values.

3.3 PDF reconciliation

Where the parser could not resolve an XBRL unit, we compared the value with the same column of the filed BRSR PDF. We accepted it if the PDF printed the unit and the number matched (`validated_pdf_unit_overrides.json` and two adjudication files). This route supplies 499 values, 352 of them in the nine core fields (tier 2).

3.4 Gap-fill from public reports

Where the XBRL was blank, agents searched the filed BRSR PDF, the annual report and the sustainability report. `consolidate_gapfill.py` accepts a candidate value if it has:

- a source URL and a verbatim quote;
- a numeric value converted to the standard unit;
- an FY2025-26 period, shown in the period, quote or notes;
- a status of “found” or “corrected”, and no flag for proxy, partial or estimated values.

Of 3,870 candidate rows, we accepted 1,007. The largest rejection reasons were “not found” (1,204), “field not used” (459), “not disclosed” (372), “not applicable” (324) and “concentration only” for air (317). We rejected 25 as the wrong concept or unit, a proxy, or unreliable, and 29 because the period was not FY2025-26 (`research/gapfill_QA.csv`). The accepted rows became 395 values in `inputs/gapfill_values.csv` (389 fills of a blank, 6 corrections) for 157 companies. In the nine core fields, 239 values come from this route (tier 3).

3.5 Energy corrections

For several companies the XBRL energy total differs from the filed PDF by a factor of 10^3 to 10^{12}, a unit slip such as kWh entered as GJ. `consolidate_energy.py` applies the rule of Section 3.4 to energy, water consumption and waste recovered, with one extra condition: **we use an energy table as a set**. If we cannot accept a company’s corrected total, we drop its corrected renewable and non-renewable split too, so a half-corrected table cannot look consistent. Of 256 candidate rows, we accepted 112: 74 fills and 38 corrections, the corrections covering 13 companies. Rejections: “not disclosed” (51), “not found” (48), proxy, partial or estimate (19), period not FY2025-26 (15), split dropped because the total failed (6), and no number (5) (`research/energy_QA.csv`). In the nine core fields, 41 values are corrections to the filed PDF (tier 5).

3.6 Revenue and profit

Revenue is Revenue from Operations for FY2025-26 in ₹ crore, from the company’s annual financial filing with NSE, on the same standalone or consolidated basis as its BRSR (`financial_revenue_ledger_all500.csv`). We use a revenue figure (`revenue_ok`) if it is verified for 1 April 2025 to 31 March 2026, or if it is verified for that year and its one open issue is an unstated BRSR boundary or a discontinued-operations note (both flagged). We do not use calendar-year filers, multi-period derivations, pro-forma or modelled revenues. 487 of the 500 companies have a usable revenue figure.

Profit after tax (`pat_fy2026.csv`) feeds the supplementary profit view (Section 9) and nothing else.

3.7 Confidence tiers

Table 3.2. Source tier of every reported value in the nine core fields (including reported zeros), all 500 companies.

BRSR field (CNPI key)	T1 XBRL, explicit unit	T2 unit checked vs PDF	T3 public-report fill	T4 unit rule R1/R2	T5 corrected to PDF	Total
`scope1_tco2e`	212	153	91	20	0	476
`scope2_tco2e`	212	149	92	21	0	474
`total_energy_gj`	448	0	13	0	13	474
`renewable_energy_gj`	449	0	12	0	12	473
`nonrenewable_energy_gj`	448	0	12	0	13	473
`water_withdrawal_m3`	452	14	6	0	2	474
`water_consumption_m3`	455	17	1	0	0	473
`waste_generated_tonnes`	456	11	5	0	1	473
`waste_recovered_tonnes`	463	8	7	0	0	478
All nine fields	3595	352	239	41	41	4268

Shares of the 4,268 values: tier 1, 84.2%; tier 2, 8.2%; tier 3, 5.6%; tier 4, 1.0%; tier 5, 1.0%. Climate is the least machine-readable pressure: 212 of 476 Scope 1 values are tier 1. Of the 409 ranked companies, 221 (54.0%) rest on at least one core value outside tier 1, and 85 (20.8%) on at least one public-report fill.

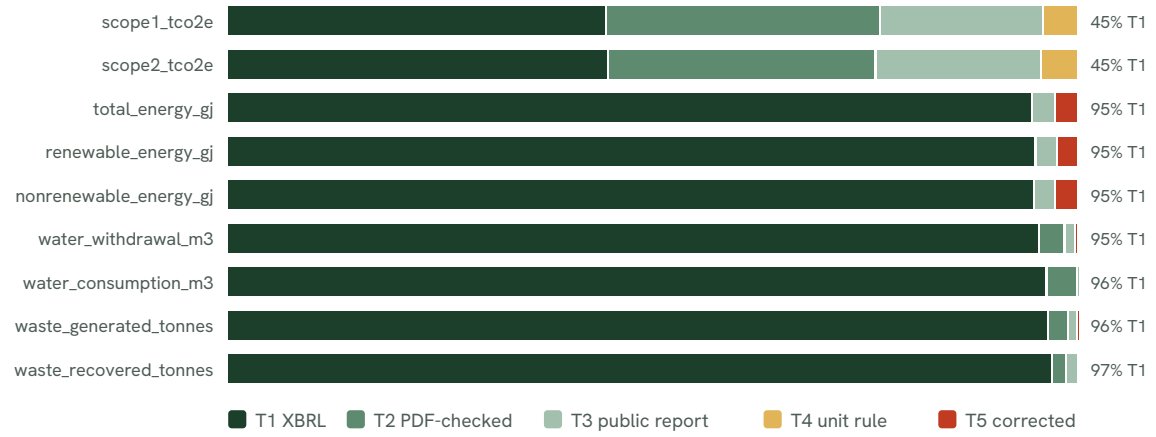


Figure 3.1. Source tier of every reported value, by BRSR field. Share of the values reported by the 500 companies (counts in Table 3.2). Tier 1 is a machine-readable XBRL value with an explicit unit.

LIMITATION.

Energy, water and waste are machine-readable for 94–97% of companies. Climate is the exception: 246 XBRL filings tag Scope 1 and 2 with the ambiguous unit "MtCO2e". The pressure that dominates the nature cost is the one that most often needed manual reconciliation.

3.8 Coverage, from reported to scored

Table 3.3. Coverage by pressure (company counts, all 500). The columns from "Zero to confirm" to "Revenue not matched" are exclusive, so usable = present − zero − held back − revenue not matched.

Pressure	Value present (of 500)	Zero to confirm	Held back (not zero)	Revenue not matched	Usable intensity	Scored in a peer group	Counts in a published rank
Climate	473	13	6	4	450	437	409
Energy	473	2	7	2	462	447	409
Water	473	6	6	1	460	446	409
Pollution	472	8	17	4	443	392	366
Air (separate rank)	245	10	13	0	222	178	178

"Scored in a peer group" falls below "usable" for two reasons: a company may sit in a group too small to score, and, for pollution, in one of the four groups where we do not score it.

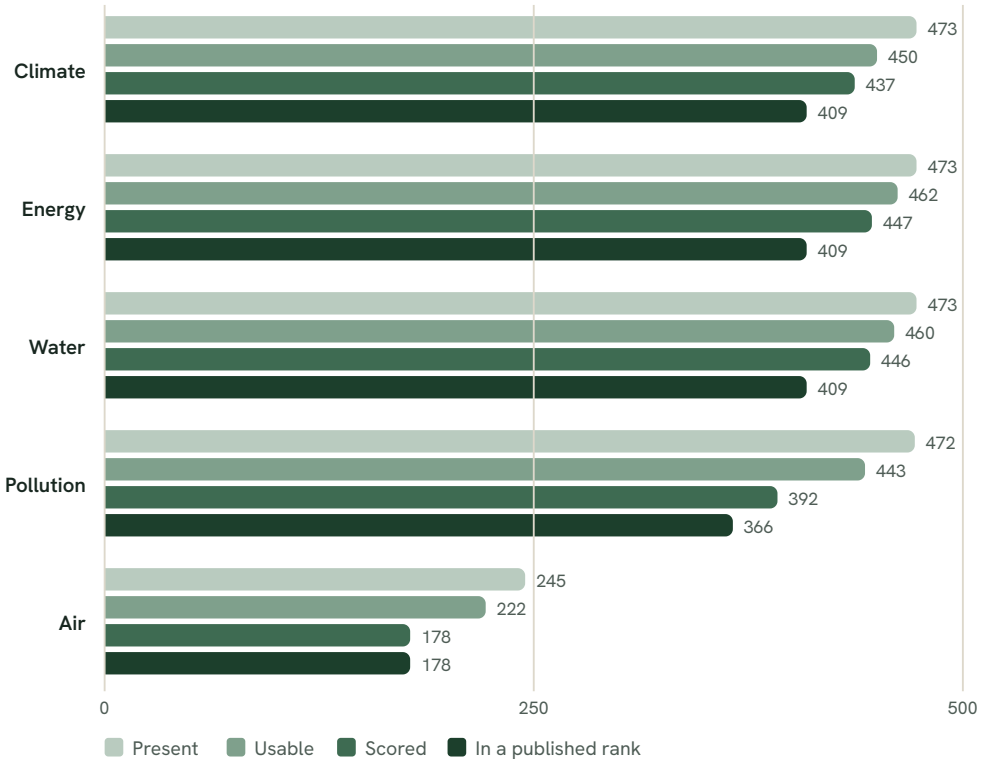


Figure 3.2. From reported to ranked, by pressure. Companies with a value present, a usable intensity, a score in a peer group, and a value that counts in a published rank (pollution is unscored in four groups; air has its own rank).

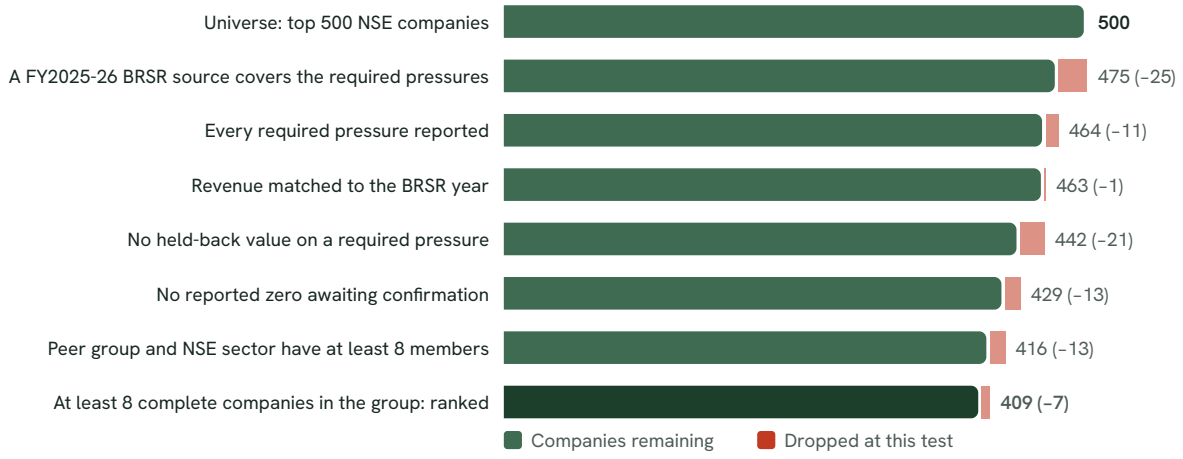


Figure 3.3. From 500 companies to 409 ranked. We count each company at the first test it fails, in data order (Section 8 gives the build's own reasons).

3.9 Land and operating sites

The BRSR has no field for land area. Our land and site evidence comes from a separate search of annual reports, mining-lease tables and environmental-clearance documents for all 500 companies, which a different AI model (DeepSeek; the "Hermes" run, `research/hermes/`) carried out on its own. Section 10 covers it.

4 Definitions of the pressures

Notation: i indexes companies, g peer groups and p pressures. Rev_i is revenue from operations in ₹ crore. All quantities are for FY2025-26 on the company's BRSR boundary.

4.1 Climate

$$C_i = S1_i + S2_i$$

where $S1_i$ is total Scope 1 emissions and $S2_i$ total Scope 2 emissions, both in tCO2e (BRSR P6 EI 7). We need both components and never use a partial sum.

- **Scope 1** covers direct emissions from fuel burnt and processes on the company's own sites. **Scope 2** covers emissions from generating the electricity, heat and steam the company buys (WRI and WBCSD 2004).
- **Scope 2 method.** The GHG Protocol asks for a location-based and a market-based figure (Sotos 2015). The BRSR asks for one number and no method. We cannot assign the method in 441 of the 465 parsed filings; 10 state market-based, 6 location-based, and 8 mention both without saying which the total uses. Dividing Scope 2 by purchased non-renewable electricity gives the factor each company applied. Across 433 companies the median is 0.710 tCO2/MWh and 286 sit at 0.71, the Central Electricity Authority's weighted-average grid factor for FY2024-25 (CEA 2025). In practice most BRSR Scope 2 figures are location-based. We use Scope 2 as reported.
- **Scope 3** is unscored. Of 465 parsed filings, 141 report it with a resolvable unit, 155 with the ambiguous "MtCO2e" unit, and 169 not at all; boundaries differ as well.
- **Double counting.** Part of one company's Scope 2 is a generator's Scope 1. You must never sum climate values across companies.

4.2 Energy

$$E_i = E_i^{\mathrm{NR}} \quad \text{and, where } E_i^{\mathrm{NR}} \text{ is not reported,} \quad E_i = \max\left(0,\ E_i^{\mathrm{T}} - E_i^{\mathrm{R}}\right)$$

where E_i^{T} is total energy consumed, E_i^{R} energy from renewable sources and E_i^{NR} energy from non-renewable sources, all in GJ (P6 EI 1). The offset is the renewable energy itself, which the pressure leaves out. The BRSR counts biomass and bagasse burnt in boilers as renewable, so these lower the energy pressure on the same terms as contracted solar or wind. The gross energy figure is E_i^{T}.

4.3 Water

Let W_i^{wd} be total water withdrawal and W_i^{c} total water consumption, both in kL (P6 EI 3). The scored water pressure is

$$W_i = \begin{cases} W_i^{\mathrm{wd}} & \text{if } 1.05\,W_i^{\mathrm{wd}} < W_i^{\mathrm{c}} \le 10\,W_i^{\mathrm{wd}} \quad \text{(capped)} \\ W_i^{\mathrm{wd}} & \text{if } W_i^{\mathrm{wd}} > 0 \text{ and } \left| W_i^{\mathrm{c}}/W_i^{\mathrm{wd}} - 0.20 \right| < 0.0005 \quad \text{(assumed discharge)} \\ W_i^{\mathrm{c}} & \text{otherwise} \end{cases}$$

We hold the value back if $W_i^{\mathrm{c}} > 10\,W_i^{\mathrm{wd}}$ (Section 6). The offset is the water discharged back, which the BRSR treats as not consumed. The gross water figure is W_i^{wd}. Section 6 gives the reasons for the two special cases (rules V6 and V7).

4.4 Pollution

$$P_i = \max\left(0,\ G_i - R_i\right)$$

where G_i is total waste generated and R_i total waste recovered through recycling, re-use or other recovery, both in tonnes (P6 EI 9). The offset is the waste recovered. A company that reports recovering more waste than it generated in the year (from stock) gets $P_i = 0$; the pressure never goes negative. The gross pollution figure is G_i.

4.5 Air (supplementary)

$$A_i = \mathrm{NO}_{x,i} + \mathrm{SO}_{x,i} + \mathrm{PM}_i \quad \text{(tonnes; all three required)}$$

We rank air on its own and keep it out of the main score (Section 9). The BRSR asks for these pollutants with "please specify unit", and many companies answer with a stack concentration (mg/Nm^3), which no one can convert to tonnes. We excluded 322 air values from 111 companies because we could not verify their unit as an annual mass.

4.6 Gross and net

Each pressure has a **gross** figure, which the zero test and the extreme-value test use (Section 6), and a **net** figure, which the score uses:

Pressure	Gross figure $Q^{\mathrm{gross}}_{i,p}$	Net figure $Q_{i,p}$
Climate	C_i	C_i (no offset)
Energy	E_i^{T}	E_i
Water	W_i^{wd}	W_i
Pollution	G_i	P_i
Air	A_i	A_i

The offsets matter for the 409 ranked companies. 59 of them get at least half their energy from renewable sources (median 10.6%). The median company recovers 81.6% of its waste; 174 recover at least 90%, and 41 report recovering all of it or more. 215 have a scored water value below their withdrawal.

5 Peer groups

Principle. We compare a company with companies whose main business is of the same kind. Intensities differ by three to four orders of magnitude between sectors: median Scope 1 + 2 per ₹ crore in cement is about 5,300 times that in insurance (`insights/pressures/PRESSURES.md`). A cross-sector rank would rank sectors.

Method.

1. **Start** from the company's NSE Indices industry classification (as of 22 October 2026). The universe spans 20 NSE sectors.
2. **Verify the main activity** from the company's own BRSR Section A product lines, which give each product's NIC 2008 code (MOSPI 2008) and its share of turnover. Where these were missing or unclear, we used the annual report. The main activity is the **largest source of turnover**. Of the 500 placements, 342 cite the BRSR XBRL file as evidence. For the 487 with a numeric share, the median share of the largest line is 95.3% (174 at 100%; 31 below 50%). Each placement carries a confidence grade: high (361), medium (128) or low (11). We flag 14 companies as conglomerates.
3. **Move** a company whose main activity belongs in another NSE sector to that sector's group. We moved **54 companies** this way (for example, Godawari Power & Ispat from Capital Goods to steel; Coal India from Oil, Gas & Consumable Fuels to mining). We record each move with its evidence and source URL (`inputs/peer_groups_v12.csv`, columns `pg_evidence`, `pg_source_url`).
4. **Split** large NSE sectors that mix unlike businesses. Capital Goods becomes machinery, engineering and defence, and electrical equipment and electronics. Chemicals becomes commodity chemicals and fertilisers, and specialty chemicals and paints. Metals & Mining becomes steel and iron products, and non-ferrous metals and mining. Healthcare becomes pharmaceuticals, and hospitals and diagnostics. Automobile and Auto Components becomes automobiles, and auto components and tyres. Construction Materials becomes cement, and building products. Financial Services becomes five sub-groups: banks; lending NBFCs and housing finance; capital markets and intermediaries; insurance; investment and holding companies.

The result is **29 peer groups**.

Minimum sizes. Three thresholds apply (all in `P`):

- the peer group, and the company's NSE sector after any move, must each have at least 8 member companies (`min_sector_size = 8`);
- we score a pressure in a group if at least 6 of its companies have a usable value for it (`min_element_reporters = 6`);
- we publish ranks if at least 8 companies in the group have a complete, clean score (`min_ranked_to_publish = 8`).

Five groups fail: building products (4 members), diversified conglomerates (1), investment and holding companies (6), textiles and apparel (7), and telecom and media (8 members, 7 of them with complete clean data). We list their 26 companies with scores where available and do not rank them.

Table 5.1. Peer groups. Pressures scored: C climate, E energy, W water, P pollution.

Peer group	Companies	Ranked	Pressures scored	Air-ranked	Profit-ranked
Healthcare: pharmaceuticals	40	40	C, E, W, P	26	38
Financial Services: Lending NBFC and housing finance	33	29	C, E, W, P	–	28
Auto components and tyres	29	27	C, E, W, P	15	27
FMCG	28	27	C, E, W, P	22	26
Capital goods: machinery, engineering and defence	31	25	C, E, W, P	11	25
Financial Services: Banks	27	25	C, E, W, P	–	25
Capital goods: electrical equipment and electronics	29	24	C, E, W, P	10	24
IT services and software	24	20	C, E, W, P	12	20

Peer group	Companies	Ranked	Pressures scored	Air-ranked	Profit-ranked
Retail, e-commerce, hotels and consumer services	27	19	C, E, W, P	9	17
Chemicals: commodity chemicals and fertilisers	18	17	C, E, W, P	13	14
Chemicals: specialty, paints and coatings	17	15	C, E, W, P	13	15
Metals: steel and iron products	17	15	C, E, W, P	9	15
Power	17	13	C, E, W	8	13
Oil, gas and fuels	15	13	C, E, W, P	8	12
Financial Services: Capital markets and financial intermediaries	18	12	C, E, W, P	–	12
Automobiles	14	12	C, E, W, P	12	10
Construction and engineering	13	12	C, E, W	–	12
Cement	10	10	C, E, W, P	10	9
Consumer durables	10	10	C, E, W, P	–	9
Metals: non-ferrous and mining	13	9	C, E, W	–	9
Financial Services: Insurance underwriting	12	9	C, E, W, P	–	–
Real estate	12	9	C, E, W	–	9
Healthcare: hospitals and diagnostics	10	9	C, E, W, P	–	9
Transport and logistics	10	8	C, E, W, P	–	–
Telecom and media	8	0	not ranked	–	–
Textiles and apparel	7	0	not ranked	–	–
Financial Services: Investment and holding companies	6	0	not ranked	–	–
Building products	4	0	not ranked	–	–
Diversified conglomerates	1	0	not ranked	–	–
Total (29 groups)	**500**	**409**		**178**	**378**

WHAT THIS MEANS.

On every pressure, peer-group medians differ by three to four orders of magnitude. A single league table of all 500 companies would rank sectors. Within a group the spread narrows, and you compare businesses that make similar things.

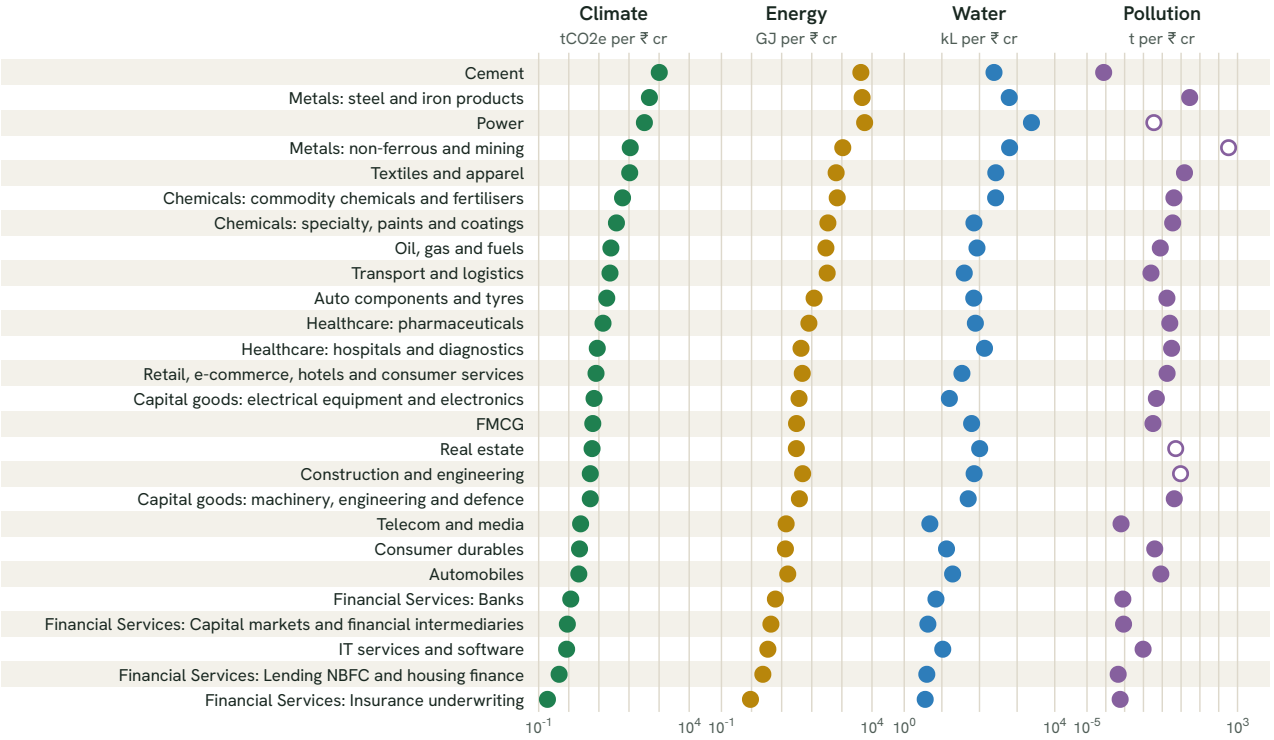


Figure 5.1. Peer-group medians by pressure. Median net intensity per ₹ crore of revenue by peer group (logarithmic scales; each gridline is a factor of ten). Groups with at least 6 usable values. Hollow markers: pollution shown but unscored in that group.

6 Data validation rules

We hold back a value that looks wrong, even if the filing prints it, until the company confirms it. A company with such a value on a pressure that counts in its group loses its rank ("rank withheld"). `build_clbi.py` holds every rule below, with its threshold in the parameter dictionary `P`. `CLBI_v1_data_queries.csv` lists all held-back values with their reasons (53 rows).

V1. Reported zeros. We flag a zero *gross* figure, or a zero Scope 1 or Scope 2 on its own, as "to confirm" and do not score it (`exclude_reported_zero`). For an operating company, a zero most often means "not measured". A zero *net* figure with a positive gross figure, such as all waste recovered, is a genuine offset, and we score it. Zero flags in this build: climate 13 companies, energy 2, water 6, pollution 8, air 10.

V2. Zero Scope 2 accepted where the renewable share is at least 90% (new in v1.2). V1 does not flag a zero Scope 2 if the company's own filing shows renewable energy at 90% or more of total energy, or renewable electricity at 90% or more of electricity consumed (`scope2_zero_re_min = 0.90`):

$$S2_i = 0 \;\wedge\; \left(\frac{E_i^{\mathrm{R}}}{E_i^{\mathrm{T}}} \geq 0.90 \;\vee\; \frac{\mathrm{El}_i^{\mathrm{R}}}{\mathrm{El}_i^{\mathrm{T}}} \geq 0.90 \right) \;\Rightarrow\; \text{accepted}$$

where $\mathrm{El}_i^{\mathrm{R}}$ and $\mathrm{El}_i^{\mathrm{T}}$ are renewable and total electricity consumed. *Rationale:* at that share, a zero fits a market-based Scope 2 on contracted renewable power. *Effect:* we accept 7 of the 11 zero Scope 2 values (HINDUNILVR, BEL, POWERINDIA, BOSCHLTD, GVT&D, UBL, CUPID).

V3. Extreme values: the 300× rule. For each pressure, we compare the gross intensity with the median of the peer group:

$$x_{i,p} = \frac{Q_{i,p}^{\mathrm{gross}}}{\mathrm{Rev}_i} \quad \left(Q_{i,p}^{\mathrm{gross}} > 0 \right), \qquad r_{i,p} = \frac{x_{i,p}}{\mathrm{median}_{j \in g}\; x_{j,p}}, \qquad \text{held if } r_{i,p} > 300 \;\text{ or }\; r_{i,p} < \tfrac{1}{300}$$

The median covers all companies in the group with a positive gross figure and usable revenue (`extreme_ratio = 300`). *Rationale:* a value this far from like-for-like peers tends to be a unit slip (a factor of 1,000 between kWh and MWh, or kg and t) or a boundary problem (head office only). We test the **gross** figure because a very low net figure can be a genuine offset. The threshold is a judgement, and we have not run a formal sensitivity test on it (Section 13). *Effect:* on the four core pressures we hold back 16 values (2 above, 14 below). We flag 11 more pollution values as "shown only", because their groups do not score pollution, and hold back 13 air values.

V4. Scope 2 implies an electricity bill larger than revenue.

$$B_i = \frac{S2_i}{0.71\ \mathrm{tCO_2/MWh}} \times 1{,}000\ \tfrac{\mathrm{kWh}}{\mathrm{MWh}} \times ₹6/\mathrm{kWh} \times 10^{-7}\ \tfrac{₹\,\mathrm{crore}}{₹}, \qquad \text{held if } B_i > \mathrm{Rev}_i$$

(`scope2_grid_ef = 0.71`, the CEA FY2024-25 weighted-average factor; `scope2_tariff_inr_per_kwh = 6.0`). *Rationale:* a company cannot pay more for grid electricity than it earns, so such a value is almost always a unit error. *Effect:* 2 companies (SHYAMMETL, PFOCUS).

V5. Scope 2 implies more grid electricity than 1.5× total energy (new in v1.2). We compare the grid electricity implied by Scope 2, in GJ, with the company's total reported energy:

$$G_i^{\mathrm{S2}} = \frac{S2_i}{0.71} \times 3.6\ \tfrac{\mathrm{GJ}}{\mathrm{MWh}}, \qquad \text{held if } G_i^{\mathrm{S2}} > 1.5\, E_i^{\mathrm{T}} \;\wedge\; \neg \mathrm{V4} \;\wedge\; \tfrac{1}{300} \leq \frac{E_i^{\mathrm{T}}/\mathrm{Rev}_i}{\mathrm{median}_{j \in g}\left(E_j^{\mathrm{T}}/\mathrm{Rev}_j \right)} \leq 300 \;\wedge\; \neg \mathrm{V10}$$

(`scope2_energy_ratio_max = 1.5`). *Rationale:* grid electricity is part of total energy consumed, so Scope 2 cannot imply much more grid electricity than all the energy the company used; the margin of 1.5 allows for a different grid factor. **We run the test against a credible energy total:** one within the 300× band of its peers that passes V10. A tiny, implausible energy total would otherwise make a normal Scope 2 look inconsistent. *Effect:* 7 companies exceed 1.5×. One already fails V4, and we do not test three whose energy total is itself held back (THERMAX, KPITTECH, GALLANTT). We hold three: COALINDIA, CANHLIFE and INGERRAND.

V6. Water consumption above withdrawal. When a company reports consumption more than 5% above withdrawal, but no more than 10 times it, it has in most cases counted recycled water as consumed. Freshwater drawn is the upper bound, so we score withdrawal (the "capped" case in Section 4.3). *Effect:* 15 companies (all ranked), with consumption 1.05 to 4.12 times withdrawal. Above **10 times** withdrawal, we hold the value as a probable unit error. *Effect:* 4 companies held (BAJFINANCE and BAJAJFINSV, which report zero withdrawal; BEML; PRIVISCL).

V7. Consumption at 20% of withdrawal: scored on withdrawal (new in v1.2). When reported consumption equals 20% of withdrawal to within 0.05 percentage points (`assumed_consumption_share = 0.20`), the company has applied an assumed 80% discharge in place of a measurement. We credit no offset, score water on withdrawal, and set the flag `water_discharge_assumed`. *Effect:* 30 companies, 24 of them ranked; most are banks, lenders and other office-based businesses. We decided on 28 October to keep the rule. Its narrow reach is a limitation (Section 13): it catches a ratio of 0.20 and no other round ratio, and no one has read the 30 filings for the word "assumed".

V8. Waste recovered more than 1.5× waste generated. We hold the value as an inconsistent table, or a stock movement too large to count as current-year recovery. *Effect:* 4 companies (TVSMOTOR, METROBRAND, IGL, PRIVISCL). We accept recovery between 1.0 and 1.5 times generation and set $P_i = 0$.

V9. Pollution unscored in four groups. In power, non-ferrous metals and mining, construction and engineering, and real estate, some companies put their largest waste streams (fly ash; overburden and tailings; construction and demolition debris) inside the BRSR waste table and others leave them out (`pollution_not_comparable`). The filings show the spread. Coal India reports 5,345,920,956 t generated, 91.3% of all waste reported by the 500 companies, and 5,345,918,279 t recovered, a net of about 2,676 t. GMDC, a lignite miner, reports 69 t generated. NTPC reports 57,193 t and Reliance Power 106,741,115 t. Larsen & Toubro reports 596,194 t against 9.7 t for IRCON and 24 t for NBCC. In real estate, Oberoi Realty reports 333,517.467 t generated and 333,517.46 t recovered, a net of 0.007 t. A waste score in these groups would reward whoever left out, or offset in full, the largest stream. We show pollution for these groups, unscored, with the reason, and rank them on climate, energy and water. We added real estate after the independent review (Appendix F).

V10. Non-renewable energy above total energy. We hold the value if $E_i^{\mathrm{NR}} > 1.01\, E_i^{\mathrm{T}}$. *Effect:* none in this build.

V11. Revenue match. A company without a usable revenue figure (Section 3.6) has no intensity and no rank. *Effect:* 13 companies.

V12. Scope 1 larger than fuel use could emit (new; a caution, not a hold). Burning fuel emits at most about 0.1 tCO2 per GJ. Outside the groups where we expect process emissions (`process_emission_groups`: cement; steel and iron products; non-ferrous metals and mining; commodity chemicals and fertilisers; oil, gas and fuels), we flag a company if

$$\frac{S1_i}{E_i} > 0.12\ \mathrm{tCO_2/GJ}$$

where E_i is its non-renewable energy (`scope1_fuel_max_t_per_gj = 0.12`; column `scope1_fuel_check`). We apply the check where both the climate value and the energy value are usable (not missing, zero-flagged or held back). *Rationale:* a higher ratio needs process or fugitive emissions that the filing does not label, or points to a unit or boundary error. *Effect:* 15 companies, 14 of them ranked. Three of the seven zero Scope 2 values accepted under V2 belong to companies that fail this check (BEL, POWERINDIA, CUPID); they keep their rank and carry a caution.

V13. Naming caution. This flag leaves the rank alone. It tells you not to single the company out by name, for example as a sector leader, until someone checks it. Three sources set it: (a) a reviewed note on the company's main activity in `inputs/peer_fit_notes.csv`, where the business differs from its peers or its BRSR boundary is unusual (for example steel-pipe converters that buy their steel, jewellers among appliance makers, a delivery platform among hotels and stores, drug makers with very low Scope 1, and UPL's standalone boundary); (b) a scored gross intensity 100–300 times above or below the peer median, inside the hold-back line but unusual; (c) rule V12. We retired an earlier rule that read the NIC division of the dominant product

as “trading” or “services”, because it mislabelled refiners and manufacturers. *Effect:* 31 ranked companies carry a caution: 12 from reviewed notes, 7 from value flags (10 flags: climate 2, energy 3, water 2, pollution 3) and 14 from V12, with some companies in more than one.

Table 6.1. Values held back or flagged on the four core pressures (`CLBI_v1_data_queries.csv`). “× median” is the gross intensity relative to the peer median.

Company	Peer group	Pressure	Test	× median	Status
PFOCUS	Real estate	climate	Scope 2 bill > revenue	13,174.4	held back
SHYAMMETL	Metals: steel and iron products	climate	Scope 2 bill > revenue	8.6	held back
CANHLIFE	Financial Services: Insurance underwriting	climate	Scope 2 grid electricity > 1.5× total energy	1/1	held back
COALINDIA	Metals: non-ferrous and mining	climate	Scope 2 grid electricity > 1.5× total energy	1.7	held back
INGERRAND	Capital goods: machinery, engineering and defence	climate	Scope 2 grid electricity > 1.5× total energy	1/1	held back
ABSLAMC	Financial Services: Capital markets and financial intermediaries	climate	gross < 1/300 of peer median	1/5,000	held back
ETERNAL	Retail, e-commerce, hotels and consumer services	climate	gross < 1/300 of peer median	1/833	held back
ABREL	Real estate	climate	gross > 300× peer median	345.1	held back
CENTRALBK	Financial Services: Banks	energy	gross < 1/300 of peer median	≈0	held back
CHOICEIN	Financial Services: Capital markets and financial intermediaries	energy	gross < 1/300 of peer median	≈0	held back
ETERNAL	Retail, e-commerce, hotels and consumer services	energy	gross < 1/300 of peer median	1/1,111	held back
GALLANTT	Metals: steel and iron products	energy	gross < 1/300 of peer median	≈0	held back
KPITTECH	IT services and software	energy	gross < 1/300 of peer median	≈0	held back
THERMAX	Capital goods: machinery, engineering and defence	energy	gross < 1/300 of peer median	≈0	held back
ABREL	Real estate	energy	gross > 300× peer median	813.6	held back
ANANTRAJ	Real estate	waste	gross < 1/300 of peer median	≈0	shown only
CONCOR	Transport and logistics	waste	gross < 1/300 of peer median	1/2,500	held back
ETERNAL	Retail, e-commerce, hotels and consumer services	waste	gross < 1/300 of peer median	1/1,111	held back
GMDCLTD	Metals: non-ferrous and mining	waste	gross < 1/300 of peer median	1/10,000	shown only
IRCON	Construction and engineering	waste	gross < 1/300 of peer median	1/1,429	shown only
JAINREC	Metals: non-ferrous and mining	waste	gross < 1/300 of peer median	1/588	shown only
NBCC	Construction and engineering	waste	gross < 1/300 of peer median	1/667	shown only
RVNL	Construction and engineering	waste	gross < 1/300 of peer median	1/1,000	shown only
ADANIPOWER	Power	waste	gross > 300× peer median	506.2	shown only
JPPOWER	Power	waste	gross > 300× peer median	813.6	shown only

Company	Peer group	Pressure	Test	× median	Status
JSWENERGY	Power	waste	gross > 300× peer median	517.3	shown only
NLCINDIA	Power	waste	gross > 300× peer median	426.8	shown only
RPOWER	Power	waste	gross > 300× peer median	22,907.6	shown only
IGL	Oil, gas and fuels	waste	recovered > 1.5× generated	1/7	held back
METROBRAND	Retail, e-commerce, hotels and consumer services	waste	recovered > 1.5× generated	1/23	held back
PRIVISCL	Chemicals: specialty, paints and coatings	waste	recovered > 1.5× generated	1.9	held back
TVSMOTOR	Automobiles	waste	recovered > 1.5× generated	1/2	held back
BAJAJFINSV	Financial Services: Lending NBFC and housing finance	water	consumption > 10× withdrawal	–	held back
BAJFINANCE	Financial Services: Lending NBFC and housing finance	water	consumption > 10× withdrawal	–	held back
BEML	Capital goods: machinery, engineering and defence	water	consumption > 10× withdrawal	1.8	held back
PRIVISCL	Chemicals: specialty, paints and coatings	water	consumption > 10× withdrawal	2.0	held back
ADANIENSOL	Power	water	gross < 1/300 of peer median	1/2,000	held back
ETERNAL	Retail, e-commerce, hotels and consumer services	water	gross < 1/300 of peer median	1/370	held back
J&KBANK	Financial Services: Banks	water	gross < 1/300 of peer median	1/769	held back
JAINREC	Metals: non-ferrous and mining	water	gross < 1/300 of peer median	1/455	held back

LIMITATION.

The hold-back rules catch values far from peers or impossible on their face. They miss some inconsistencies inside a filing. A simple cross-table test, Scope 2 divided by purchased non-renewable electricity, falls outside 0.5–1.0 tCO2/MWh for 42 companies, 34 of them ranked, and no rule above holds any of them back (Section 13).

7 Scoring

7.1 Intensity

For each company and pressure,

$$I_{i,p} = \frac{Q_{i,p}}{\mathrm{Rev}_i}$$

in units per ₹ crore. $I_{i,p}$ is undefined if $Q_{i,p}$ is missing, zero-flagged (V1) or held back (V3–V8, V10), or if revenue is unusable (V11).

7.2 Percentile score

Within peer group g, let $\mathcal{V}_{g,p}$ be the set of companies with a defined $I_{i,p}$, and $k = |\mathcal{V}_{g,p}|$. We score the pressure if $k \geq 6$. Let $\rho_{i,p}$ be the position of company i when we sort the intensities in $\mathcal{V}_{g,p}$ from lowest to highest, with tied values sharing the average of their positions. The score is

$$s_{i,p} = 100 \times \frac{k - \rho_{i,p}}{k - 1}$$

The lowest intensity scores 100, the highest 0, and the others sit at even steps by position. The score uses the order and ignores the size of the gaps, which keeps it stable against the long right tails all four pressures show.

$\mathcal{V}_{g,p}$ includes every company with a usable value for pressure p, including companies whose rank we withhold because of a problem on a *different* pressure. The energy score covers more companies than we rank in 15 of the 24 ranked groups (climate and water: 14 groups; pollution: 9).

7.3 Averaging

Let $\mathcal{R}_g$ be the required pressures of group g: all four, or climate, energy and water in the four groups of rule V9. The final score is the unweighted mean

$$F_i = \frac{1}{|\mathcal{R}_g|} \sum_{p \in \mathcal{R}_g} s_{i,p}$$

defined if every $s_{i,p}$ for $p \in \mathcal{R}_g$ exists and no required pressure carries an open data query. A company gains nothing by leaving a value out. Equal weights are a choice we made, with no empirical basis (OECD and JRC 2008); Section 12 tests what dropping each pressure does.

7.4 Ranking and rounding

We round scores to six decimal places before ranking, so floating-point noise cannot split a genuine tie. Let $\mathcal{K}_g$ be the companies in g with a defined F_i. Then

$$\mathrm{rank}_i = 1 + \left|\{\, j \in \mathcal{K}_g : \tilde{F}_j > \tilde{F}_i \,\}\right|, \qquad \tilde{F} = \mathrm{round}\ (\ F, 6\ )$$

This is standard competition ranking (1, 2, 2, 4). We publish ranks if $|\mathcal{K}_g| \geq 8$. In this build 30 ranked companies, in 6 groups, share a rank with at least one peer. We assign quartiles from $\mathrm{rank}_i / |\mathcal{K}_g|$ with cut-points 0.25, 0.50 and 0.75. Reports show scores to one decimal place; the files keep full precision.

7.5 Worked example: ACC Ltd (cement)

ACC reports on a standalone basis. Its FY2025-26 revenue from operations is ₹25,766.48 crore. All ten cement companies have complete, clean data, so $k = 10$ for every pressure and $k - 1 = 9$.

Step 1: net quantities (from `CLBI_v1_scores_all500.csv`).

- Climate: $C = 14,737,570 + 555,799 = 15,293,369$ tCO2e.
- Energy: $E = 71,419,552 - 5,355,975 = 66,063,577$ GJ.
- Water: withdrawal and consumption are both 4,141,552 kL; neither special case applies, so $W = 4,141,552$ kL.
- Pollution: $P = 398,506.78 - 398,506.00 = 0.78$ t.

Step 2: intensities. Dividing by 25,766.48 gives 593.54 tCO2e, 2,563.93 GJ, 160.73 kL and 0.0000303 t per ₹ crore.

Step 3: positions among the ten cement companies.

Pressure	ACC intensity (per ₹ cr)	Lower peers	Position ρ	Score $100\,(\,10-\rho\,)\,/9$
Climate	593.54	JSW Cement 554.54	2	88.89
Energy	2,563.93	none (JSW Cement next, 2,626.94)	1	100.00
Water	160.73	Shree Cement 144.45; Ambuja 146.37	3	77.78
Pollution	0.0000303	J.K. Cement 0; Ramco 0 (tied, positions 1 and 2, each 1.5)	3	77.78

J.K. Cement and Ramco Cements each report recovering more waste than they generated in the year (10,013.76 t against 9,677.51 t, and 99,098 t against 98,128.48 t). Their net is zero, and they share position 1.5, scoring $100 \times\,(\,10-1.5\,)\,/9 = 94.44$.

Step 4: average. $F = \,(\,88.89 + 100 + 77.78 + 77.78\,)\,/4 = 86.11$.

Step 5: rank. The next-highest cement score is JSW Cement's 80.56, so ACC is **rank 1 of 10.** On the supplementary views (Section 9), ACC is 2nd of 10 per tonne of cement (28,970,000 t) and 2nd of 10 on air.

8 Exclusion lists

For every peer group, we list each company we do not rank, with a plain reason and the values behind it (`exclusions_v12.py`, writing `CLBI_v1_exclusions.csv` and `CLBI_v1_exclusion_summary.csv`). Each company gets one category, tested in this order:

1. **Held back: implausible or inconsistent value.** The company has a held-back value (V3–V8, V10) on a pressure that counts in its group; the detail names the pressure, the test and the multiple of the peer median.
2. **Reported zero awaiting confirmation** (V1).
3. **Pressure not reported.**
4. **Revenue year or boundary does not match the BRSR** (V11).
5. **Peer group too small to rank** (Section 5).

Table 8.1. Companies not ranked, by category.

Category	Companies
Held back: implausible or inconsistent value	24
Peer group too small to rank	25
Pressure not reported	16
Revenue year or boundary does not match the BRSR	13
Reported zero awaiting confirmation	13
Total not ranked	**91**

Appendix C gives the same counts for each peer group.

A second view of the same 91. If we assign each company to the *first* stage it fails, in data order, the counts are: no FY2025-26 BRSR source that could complete the required pressures, 25; a required pressure missing, 11; revenue not matched, 1; a held-back value, 21; a zero to confirm, 13; group or NSE sector below 8 members, 13; fewer than 8 complete companies in the group, 7. A missing XBRL file did not drop a company on its own. Of the 35 companies without one, we completed 10 from filed PDFs or public reports, and 7 of these are ranked (HYUNDAI, BPCL, HINDPETRO, PIRAMALFIN, ITI, KIOCL, HBLENGINE).

9 Supplementary views

These views leave the main rank unchanged.

Air rank. In groups where at least 8 companies with usable revenue report a usable NOx + SOx + PM mass, we score air with the formula of Section 7.2 and rank it. We rank **178 companies in 14 groups** on air. Many companies report stack concentrations, which no one can convert to tonnes, and 10 companies with more than 1 Mt of Scope 1 report no usable air tonnage.

Output yardsticks: per tonne and per MWh. Revenue favours companies that sell at a high price. We adopted a physical yardstick where it is recognised, often disclosed and hard to redefine. Of 23 groups and sub-groups tested, two passed (`research/comparator_table.csv`): **cement**, per tonne of cement produced, and **power**, per MWh generated. For each, we divide the core pressures by the output figure, then score and average them in the same way, with the zero and hold-back flags of Section 6. We rank ten cement companies and ten power generators this way. The Spearman correlation (Spearman 1904) with the revenue rank is 0.88 for cement and 0.85 for power. Transmission companies have no generation figure and sit outside the power view.

Profit basis. As a check, we divide each pressure by profit after tax. We leave out loss-making companies and flag a margin below 2% as unstable (`thin_profit_margin = 0.02`). 378 companies are ranked on this basis. Of the 500, 440 have a usable profit, 24 have none loaded, 21 made a loss and 15 have a margin below 2%. Across groups, the median correlation between revenue-based and profit-based ranks is 0.82 (Table 12.1).

10 Land and operating-site evidence

10.1 Land data in the BRSR

The BRSR asks for the *number* of plants and offices (Section A), a free-text table of operations in or near ecologically sensitive areas (P6 EI 11), and environmental impact assessments (EI 12). It has no field for hectares, coordinates, land-use change or restoration. Land data must come from annual reports, mining-lease tables and environmental-clearance (EC) letters, with mixed periods and boundaries. On 28 October we decided to show land as **evidence only**, with no score and no rank, until comparable data exist.

10.2 The Hermes search

An independent run with a different AI model (DeepSeek, called "Hermes") searched all 500 companies (`research/hermes/`). It worked to six rules:

- no figure without a primary document (company filing, exchange attachment, EC letter, or `.gov.in` or `.nic.in` page);
- "partial" means a real number for one plant, one subsidiary or an older year, and never a company total;
- restoration stays separate from land held;
- approximate figures ("~") carry no hectare value;
- coordinates appear where an EC document printed them, and nothing is geocoded;
- conflicts between an EC letter and a filing are recorded, never settled by choice.

The run produced 590 company rows (statuses: not disclosed in area 313; partial only 165; found 93; not found 19) and 326 site rows, and kept 710 source PDFs. `land_evidence_v12.py` merges these into one category per company. A first pass assigned categories from the period and boundary text. After the independent review (Appendix F), we **classified by hand** the 28 companies that pass had placed in its top two categories, reading their quotes (`inputs/land_category_overrides.csv`). We separated operational land held from real-estate land banks, single sites, leases, concessions and allotments. We corrected two figures to company totals from the same quotes: BEL, 3,075.49 acres = 1,244.6 ha; BHEL, 16,690.71 acres = 6,754.5 ha.

Table 10.1. Land-evidence categories (`CLBI_v1_land_evidence.csv`).

Category	Definition	Companies
A	Company-wide land held (total, or owned plus leased), checked by hand	13
R	Real-estate land bank (includes joint-development land)	6
B	Figure covers a different or partial boundary: a single site, a mining or petroleum lease, a subsidiary's concession, or a group boundary	7
C	Partial or older figure only, or a statement that is not land held (an allotment, a land-acquisition claim)	51
D	Offices only, or land not disclosed as an area	115
E	Land not disclosed as an area	308
	Total	**500**

The 13 category A companies are BEL, BHEL, HAL, JSW Steel, Tech Mahindra, Vedanta, FACT, Shyam Metalics, Emmvee, Acutaas, NMDC Steel, JSW Infrastructure and Bharat Coking Coal; the last figure is for FY2024-25.

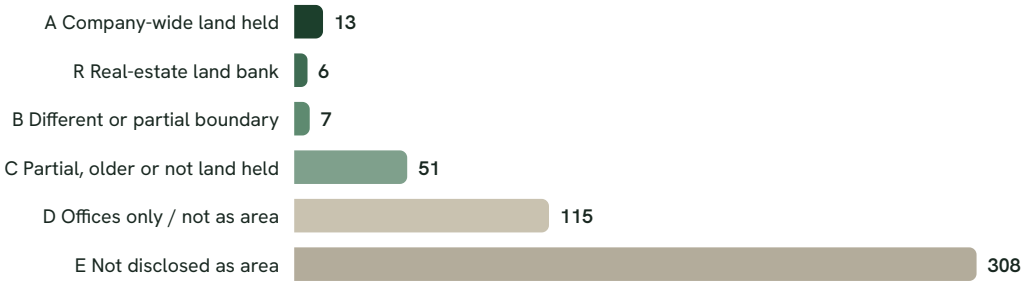


Figure 10.1. Land disclosure across the 500 companies (categories of Table 10.1).

Sites. 187 companies name at least one site, and 140 give an area for at least one (252 sites, median 21.0 ha). 9 companies print coordinates, for 10 sites. The "near a sensitive area" field holds 87 statements from 59 companies. On reading, 49 are denials and 20 are placeholders. That leaves 18 affirmative statements from 14 companies, none with coordinates (`insights/pressures/PRESSURES.md`, §6). We found a restoration area on own land for 37 companies.

Limits of the evidence. Some category A figures sit on a different boundary from the BRSR. JSW Steel's covers its India operations including subsidiaries, against a standalone BRSR; Tech Mahindra's covers 13 Indian locations, against a consolidated BRSR. Field types differ too: a petroleum mining lease (Oil India, 482,884 ha) or a subsidiary's concession (NAVA, 36,000 ha) is a licence area, so both sit in category B. We publish no ratio of land to revenue.

10.3 Proposed future biodiversity metric

Once site data allow it, we propose a site-level pressure:

$$\mathrm{BP}_i = \sum_{s \in \mathcal{S}_i} a_s \, (1 - \mathrm{MSA}_{u(s)}) \, w_s \; - \; \sum_{s \in \mathcal{S}_i} a_s^{\mathrm{rest}} \, \Delta\mathrm{MSA}_s$$

where:

- $\mathcal{S}_i$ is the set of operating sites of company i on its BRSR boundary;
- a_s is the area of site s in hectares;
- $u(s)$ is the land use of the site (for example open-cast mine, industrial plant, greenbelt);
- MSA_u is the mean species abundance under that land use: the mean abundance of the original species relative to an undisturbed state, from the GLOBIO model (Alkemade et al. 2009; Schipper et al. 2020). $1 - \mathrm{MSA}$ is the share of original abundance lost;
- $w_s \geq 1$ is a proximity weight that rises for sites inside or near protected areas, Key Biodiversity Areas, Ramsar sites, tiger reserves or eco-sensitive zones. An overlay of site polygons on the World Database on Protected Areas (UNEP-WCMC and IUCN 2026) and the KBA layer (IUCN 2016) would set it; we have not yet chosen its functional form;
- a_s^{rest} and $\Delta\mathrm{MSA}_s$ are the area restored on the company's own land and the MSA gain it achieves, so restoration offsets the same pressure, as the other offsets do.

BP_i would be in MSA-loss hectare-equivalents, divided by revenue like the other pressures. **We cannot compute it today.** It needs site areas and coordinates (or polygons, as SBTN requires for sites of 10 ha or more) for every material site, and coordinates exist for 10 of 326 named sites.

> LIMITATION.
> Land-use change is the largest direct driver of biodiversity loss on land (IPBES 2019), and the index cannot yet measure it. Until companies report site coordinates and areas, a land or biodiversity score would rest on the 13 companies (2.6%) that publish a company-wide figure for land held.

11 Illustrative nature cost

Status. This is the second product in this paper, and we keep it apart from the first. It is illustrative and never enters a rank. It puts a price on reported pressure; it is no liability, expense, bill or valuation of any company.

11.1 Method

For each company and each priced pressure, we multiply the scored net quantity by a damage price in constant 2025 rupees:

$$\mathrm{NC}_{i,p} = \frac{q_{i,p}\,\pi_p}{10^7} \quad (\text{₹ crore}), \qquad \mathrm{NC}_i = \sum_p \mathrm{NC}_{i,p}$$

where $q_{i,p}$ is the quantity as scored in Section 4 (climate C_i; energy E_i; water W_i; pollution P_i; NOx, SOx and PM each on its own) and π_p is the price in ₹ per unit. The **headline metric** is the cost as a share of EBITDA, with the share of revenue alongside:

$$\mathrm{NC}_i^{\mathrm{EBITDA}} = 100 \times \frac{\mathrm{NC}_i}{\mathrm{EBITDA}_i}, \qquad \mathrm{NC}_i^{\mathrm{Rev}} = 100 \times \frac{\mathrm{NC}_i}{\mathrm{Rev}_i}$$

EBITDA is the FY2025-26 figure in `pat_fy2026.csv`. We compute the EBITDA share for every non-financial company with positive EBITDA (370 companies), and all peer-group medians use this basis. We flag thin-margin companies and keep them in. EBITDA does not suit banks, lenders and insurers (96 companies); it is not loaded for 12 companies and is zero or negative for 6. Where EBITDA is below 5% of revenue we flag the share as unstable (8 companies: FACT, PATANJALI, RADICO, UBL, ITI, AWL, REDINGTON, BASF).

Presentation. We report the EBITDA share as **peer-group medians with their interquartile range**. We show a company value with its flags and its low–high range, because a single questionable input, such as a Scope 1 figure flagged under rule V12, can dominate one value. The rules (`v11_extensions.py`, `nature_cost_block`; prices in `inputs/prices_v12.csv`):

- we do not cost a held-back value (Section 6);
- we show a company total where climate is costed, so totals compare;
- low and high totals use all low and all high prices;
- **we report per company or as peer-group medians.** We publish no sector or index total: the buyer and the generator both count Scope 2 and grid electricity, so sums across companies double-count;
- land is evidence only, and we do not cost it.

11.2 Prices

Table 11.1. Central price set (set B) (`inputs/prices_v12.csv`; method, conversions and sources in `insights/prices/PRICES_V12.md`).

Pressure	Unit	Low	Central	High	Basis of the central value
Climate (headline, India-borne)	₹/tCO2e	2,800	5,400	11,500	23.9% India share (median of Ricke et al. 2018, Tol 2019, Thaller et al. 2026) × US EPA (2023) global social cost of CO2 of ₹22,598
Climate, global (reference only, never added)	₹/tCO2e	13,900	22,600	38,400	US EPA (2023) at 2.0% (2.5% and 1.5% for low and high); of the order of Rennert et al. (2022)
Energy depletion	₹/GJ non-renewable	0	80	550	CE Delft (2024) external price of fossil-resource scarcity; low 0 (scarcity rent already in fuel prices); high ReCiPe 2016 surplus cost at India's fossil mix

Pressure	Unit	Low	Central	High	Basis of the central value
Water (priced by scarcity)	₹/kL scored	39.6	358	440	IFVI and VBA (2026) India value factor for water consumption, including the cost of future access, scaled to India's industrial-weighted water stress (WRI Aqueduct 4.0: 4.07 against a national 4.11; Kuzma et al. 2023). Low: damage and opportunity-cost value (Kumar 2006; IFVI ecosystem services; CE Delft). High: IFVI value scaled to the "arid" stress band
Waste not recovered	₹/t	300	600	9,200	Median of five income-adjusted landfill-damage values; high from the only India-specific estimate
SOx	₹/t	85,000	2,09,000	13,69,000	Median of four India health-damage estimates, Indian value of statistical life (unchanged from v1.1)
NOx	₹/t	1,11,000	2,10,000	9,93,000	Median of three estimates (unchanged)
PM	₹/t	1,55,000	3,06,000	9,34,000	CE Delft PM10 value transferred to India (unchanged)

Sensitivity lenses. We compute two alternatives for every company (columns `nature_cost_lens_A` and `nature_cost_lens_C`):

- **Lens A (damage-only water).** Set B, with water at ₹39.6/kL, the damage and opportunity-cost value without the future-access term.
- **Lens C (global value of life for air).** Set B, with air priced on a global value of statistical life: the IFVI Global Value Factor Database India rural factors, SOx ₹10.2 lakh, NOx ₹3.4 lakh and PM ₹15.9 lakh per tonne (as implemented in `v11_extensions.py`).

We do not apply site-level water stress. Company totals carry no location, so one national price applies to every company. The waste price rests on weaker evidence than the others: its five central sources lie within 2.6 times of each other, and the full range spans 31 times.

11.3 Results

We apply prices to 465 companies with a costed total. The headline is the peer-group median of the EBITDA share (Table 11.3 and Figure 11.1). Table 11.2 gives the median over all companies for context.

Table 11.2. Illustrative nature cost across all costed companies (context for the peer-group medians).

Measure	Lens A (water ₹39.6/kL)	Set B (central)	Lens C (air at global VSL)	Companies
% of EBITDA	3.3%	**4.4%**	4.8%	370
% of revenue	0.41%	**0.58%**	0.59%	460
Share of costed total: climate	76.3%	**68.8%**	54.4%	465
water	1.4%	**11.1%**	8.8%	465
energy depletion	10.6%	**9.5%**	7.5%	465
air	10.3%	**9.3%**	28.4%	465
waste	1.4%	**1.2%**	1.0%	465

The shares describe how the summed cost splits across pressures; the sum itself has no meaning as a total. Under set B the median company's climate share is 64.5%. Climate is the largest component for 356 of the 465 companies and water for 106; climate falls below half the total for 124. The cost exceeds EBITDA for 39 companies and revenue for 9. At the global social cost of carbon, the climate cost alone would be 4.19 times the India-borne figure (₹22,600 ÷ ₹5,400).

Seawater kept out of priced water. The water price is a freshwater value, and BRSR water consumption can include seawater. For the largest water users, we read the seawater inside reported consumption from each company's own BRSR (`insights/prices/seawater_split.csv`) and recorded it with its basis and a quote (`inputs/seawater_in_consumption.csv`): seawater withdrawn minus seawater discharged to sea, or the seawater consumption the report states. We remove it from the priced quantity for 7 companies (RELIANCE, ADANIPOWER, ONGC, BPCL, TATAPOWER, HINDPETRO, NLCINDIA), 408 million kL in all; JSWENERGY reports all its seawater returned, so we remove nothing. Where the filing is unclear (NTPC, VEDL, HINDALCO,

TATACHEM, GRASIM, MRPL) we remove nothing, and their priced water may include seawater. The adjustment changes the nature cost and leaves the ranks alone; ranks use water as scored in Section 4 (column `water_priced_kl`).

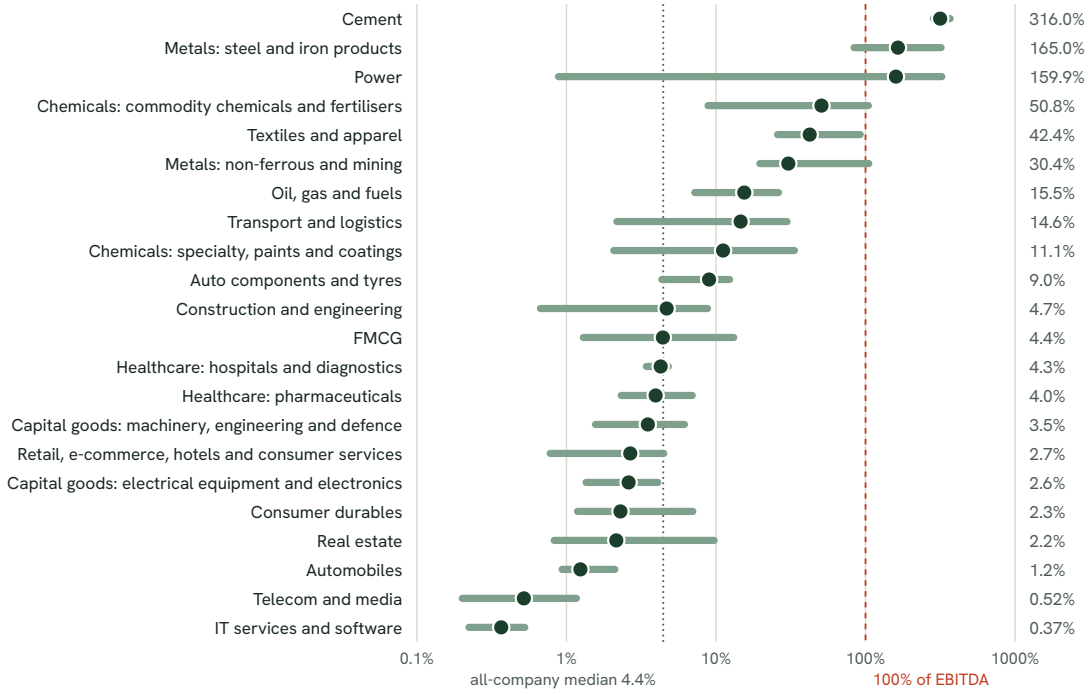


Figure 11.1. Illustrative nature cost as a share of EBITDA, by peer group (set B; median and interquartile range; logarithmic scale; non-financial groups with at least 5 companies with positive EBITDA).

Table 11.3. Illustrative nature cost by peer group, % of EBITDA (set B, with lens medians; basis: all 370 non-financial companies with positive EBITDA, thin-margin companies included and flagged; groups with at least 5).

Peer group	n	Median (set B)	IQR	Lens A	Lens C
Cement	10	316.0%	283.7%–368.2%	311.6%	334.1%
Metals: steel and iron products	16	165.0%	83.8%–320.6%	156.1%	178.3%
Power	15	159.9%	0.88%–323.6%	145.8%	219.0%
Chemicals: commodity chemicals and fertilisers	17	50.8%	8.8%–104.5%	36.4%	50.8%
Textiles and apparel	7	42.4%	25.8%–92.1%	36.6%	43.6%
Metals: non-ferrous and mining	10	30.4%	19.6%–105.1%	21.7%	34.4%
Oil, gas and fuels	13	15.5%	7.2%–26.2%	15.5%	15.5%
Transport and logistics	9	14.6%	2.2%–29.8%	14.4%	14.6%
Chemicals: specialty, paints and coatings	16	11.1%	2.1%–33.4%	9.6%	11.4%
Auto components and tyres	27	9.0%	4.3%–12.3%	7.6%	9.6%
Construction and engineering	12	4.7%	0.67%–8.8%	2.8%	4.8%
FMCG	27	4.4%	1.3%–13.1%	3.1%	5.3%
Healthcare: hospitals and diagnostics	9	4.3%	3.4%–4.8%	2.1%	4.3%
Healthcare: pharmaceuticals	40	4.0%	2.3%–6.9%	3.1%	4.3%
Capital goods: machinery, engineering and defence	28	3.5%	1.6%–6.2%	1.9%	3.6%
Retail, e-commerce, hotels and consumer services	23	2.7%	0.78%–4.5%	1.6%	2.7%
Capital goods: electrical equipment and electronics	25	2.6%	1.4%–4.1%	2.2%	2.9%
Consumer durables	10	2.3%	1.2%–7.0%	1.7%	2.3%
Real estate	10	2.2%	0.83%–9.7%	1.4%	2.2%
Automobiles	11	1.2%	0.94%–2.1%	0.86%	1.3%

Peer group	n	Median (set B)	IQR	Lens A	Lens C
Telecom and media	8	0.52%	0.20%–1.2%	0.47%	0.52%
IT services and software	22	0.37%	0.22%–0.53%	0.21%	0.37%

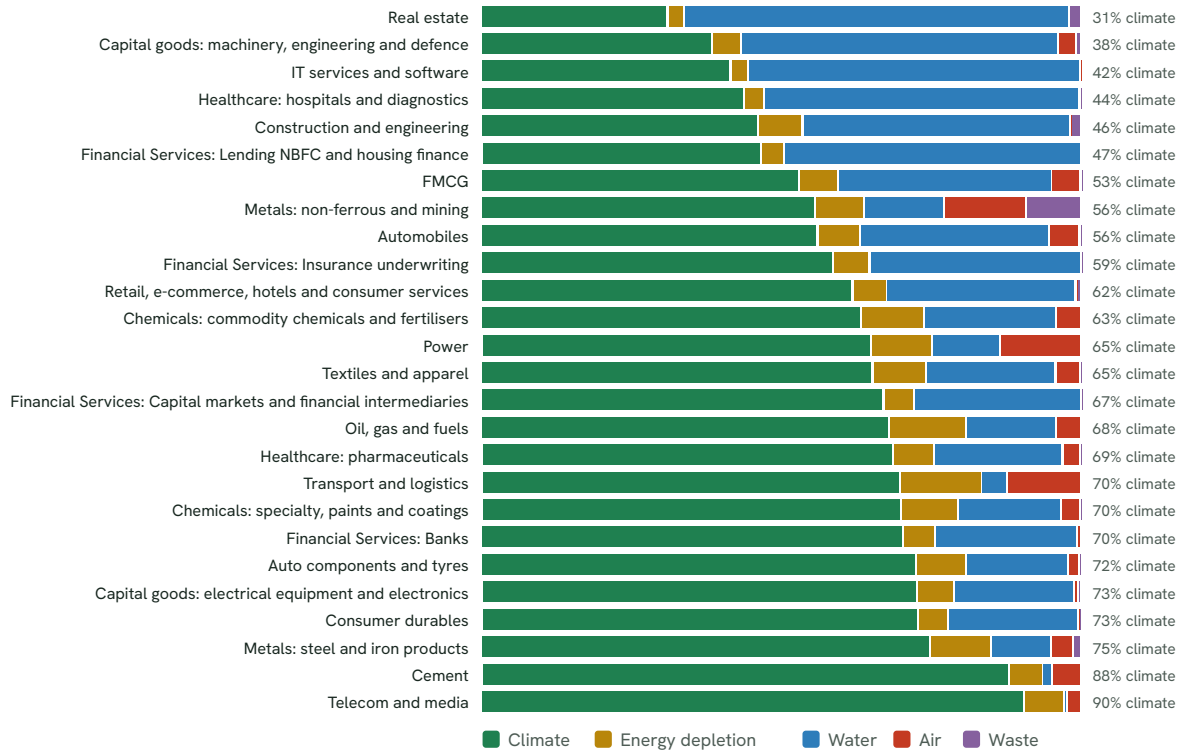


Figure 11.2. The cost by pressure, by peer group (set B). Each bar splits the group's cost by pressure: the summed cost of each pressure divided by the group's summed cost, shown as a composition.

Table 11.4. Composition by peer group (set B; groups with at least 5 costed companies; from `CLBI_v1_rankings_by_sector.xlsx`, sheet "Nature cost by sector"). Pressure shares are the median of each company's share; % of revenue is the median with the interquartile range.

Peer group	n	% of revenue	IQR	Climate	Energy	Water	Air	Waste
Cement	10	62.63	53.40–68.48	89%	6%	2%	4%	0.0%
Power	15	38.01	0.76–130.93	67%	7%	10%	12%	0.0%
Metals: steel and iron products	16	34.15	12.61–42.10	73%	10%	8%	2%	0.4%
Metals: non-ferrous and mining	10	12.88	6.19–25.78	60%	9%	16%	0%	4.6%
Textiles and apparel	7	7.04	4.27–10.59	65%	6%	20%	1%	0.1%
Chemicals: commodity chemicals and fertilisers	18	6.44	1.63–14.34	65%	9%	19%	3%	0.1%
Chemicals: specialty, paints and coatings	16	2.91	0.37–7.66	67%	8%	20%	1%	0.2%
Oil, gas and fuels	14	2.08	1.11–3.81	73%	13%	12%	1%	0.0%
Transport and logistics	9	1.61	1.40–8.18	60%	9%	15%	0%	0.0%
Auto components and tyres	27	1.42	0.80–2.00	70%	7%	20%	0%	0.1%
Healthcare: pharmaceuticals	40	1.15	0.76–1.85	66%	6%	25%	2%	0.1%
Healthcare: hospitals and diagnostics	9	1.04	0.80–1.31	48%	3%	49%	0%	0.3%
Capital goods: machinery, engineering and defence	28	0.88	0.29–1.73	55%	5%	31%	0%	0.4%
FMCG	27	0.77	0.42–1.35	52%	5%	39%	3%	0.0%
Retail, e-commerce, hotels and consumer services	22	0.64	0.27–1.78	56%	5%	39%	0%	0.1%
Construction and engineering	12	0.59	0.13–1.17	50%	7%	43%	0%	0.1%
Real estate	10	0.58	0.42–1.92	30%	3%	66%	0%	0.2%
Capital goods: electrical equipment and electronics	24	0.43	0.25–0.60	75%	6%	18%	0%	0.1%

Peer group	n	% of revenue	IQR	Climate	Energy	Water	Air	Waste
Automobiles	13	0.21	0.18–0.32	55%	6%	33%	3%	0.2%
Consumer durables	10	0.20	0.10–0.54	67%	5%	27%	0%	0.2%
Telecom and media	8	0.15	0.10–0.54	86%	7%	7%	0%	0.0%
Financial Services: Banks	26	0.10	0.08–0.13	69%	5%	25%	0%	0.0%
IT services and software	22	0.07	0.06–0.14	50%	3%	46%	0%	0.1%
Financial Services: Capital markets and financial intermediaries	15	0.07	0.04–0.14	64%	5%	30%	0%	0.0%
Financial Services: Lending NBFC and housing finance	33	0.04	0.02–0.11	52%	4%	44%	0%	0.0%
Financial Services: Insurance underwriting	11	0.03	0.01–0.03	34%	3%	63%	0%	0.0%

Worked example (ACC). Climate 15,293,369 t × ₹5,400 = ₹8,258.42 crore; energy 66,063,577 GJ × ₹80 = ₹528.51 crore; water 4,141,552 kL × ₹358 = ₹148.27 crore; pollution 0.78 t × ₹600 = ₹468; air (7,996 t NOx × ₹2,10,000) + (457 t SOx × ₹2,09,000) + (259 t PM × ₹3,06,000) = ₹185.39 crore. Central total ₹9,120.59 crore (low ₹4,395.20 crore; high ₹22,283.86 crore). With EBITDA of ₹3,306.46 crore and no flag, this is 275.8% of EBITDA (range 132.9% to 673.9% at all-low and all-high prices) and 35.4% of revenue; lens A gives ₹8,988.72 crore (271.9% of EBITDA) and lens C ₹9,294.85 crore (281.1%). The global-SCC reference for climate alone would be ₹34,563.01 crore.

WHAT THIS MEANS.

At India-borne prices, carbon makes up 68.8% of the unpriced cost, and the second pressure depends on the sector: water in services, real estate and hospitals; energy depletion and air in heavy industry. For the median company the cost is a few per cent of EBITDA. For cement, steel and coal power it exceeds EBITDA. Treat these as orders of magnitude for discussion; none of them is a liability.

PRICE CONVENTION: CHECKED.

Set B mixes conversion conventions. The water price of ₹358/kL is a global-perspective IFVI value converted at purchasing-power parity; carbon uses the market exchange rate. We decided on 28 October to keep set B. As a check, we recomputed every price from one standard (IFVI and CE Delft) and converted all of them by a single convention (`insights/prices/convention_check.csv`). The cost mix comes out the same under purchasing-power parity and under the market exchange rate: carbon 71.6–71.9%, water 15.0–15.1%, energy 2.6–2.7%, air 9.3–9.4%, waste 1.2%. The comparable set B row, before the seawater adjustment, gives carbon 66.8%, water 13.3%, energy 9.6%, air 9.1%, waste 1.2% (same file). The conclusions hold under either convention. Three caveats remain. Each price transfers published estimates, several from outside India. The IFVI water value is per kilolitre of freshwater, so we keep seawater out of the priced quantity where the filing allows (Section 11.3). The waste price rests on five sources of modest quality.

12 Robustness and sensitivity

We computed three alternative constructions for every ranked group (`CLBI_v1_robustness.csv`) and compared each with the published score by Spearman's rank correlation ρ (Spearman 1904). The composite-indicator literature recommends testing how each construction choice moves ranks (Saisana, Saltelli and Tarantola 2005; OECD and JRC 2008).

1. **Net vs gross.** We replace each pressure by its gross figure (total energy, water withdrawn, waste generated) and score and average as before. This shows how far the offsets move ranks.
2. **Leave one pressure out.** We recompute the final score four times, each time without one core pressure, and report the lowest of the four correlations.
3. **Revenue vs profit.** We compare the revenue-based final score with the profit-based one.

Table 12.1. Rank correlations by peer group.

Peer group	Ranked	ρ net vs gross	Min ρ, leave one out	ρ revenue vs profit	n (profit)
Auto components and tyres	27	0.94	0.96	0.78	27
Automobiles	12	0.82	0.90	0.84	10
Capital goods: electrical equipment and electronics	24	0.89	0.96	0.86	24
Capital goods: machinery, engineering and defence	25	0.92	0.94	0.95	25
Cement	10	0.94	0.89	0.85	9
Chemicals: commodity chemicals and fertilisers	17	0.89	0.93	0.70	14
Chemicals: specialty, paints and coatings	15	0.94	0.92	0.86	15
Construction and engineering	12	0.99	0.97	0.88	12
Consumer durables	10	0.92	0.93	0.85	9
FMCG	27	0.85	0.93	0.89	26
Financial Services: Banks	25	0.93	0.91	0.52	25
Financial Services: Capital markets and financial intermediaries	12	0.92	0.92	0.82	12
Financial Services: Insurance underwriting	9	0.98	0.95	–	0
Financial Services: Lending NBFC and housing finance	29	0.95	0.92	0.92	28
Healthcare: hospitals and diagnostics	9	0.87	0.90	0.71	9
Healthcare: pharmaceuticals	40	0.95	0.96	0.74	38
IT services and software	20	0.79	0.83	0.69	20
Metals: non-ferrous and mining	9	0.93	0.95	0.88	9
Metals: steel and iron products	15	0.97	0.97	0.82	15
Oil, gas and fuels	13	0.98	0.97	0.61	12
Power	13	0.95	0.97	0.96	13
Real estate	9	0.87	0.93	0.65	9
Retail, e-commerce, hotels and consumer services	19	0.98	0.94	0.64	17
Transport and logistics	8	0.90	0.81	0.43	7

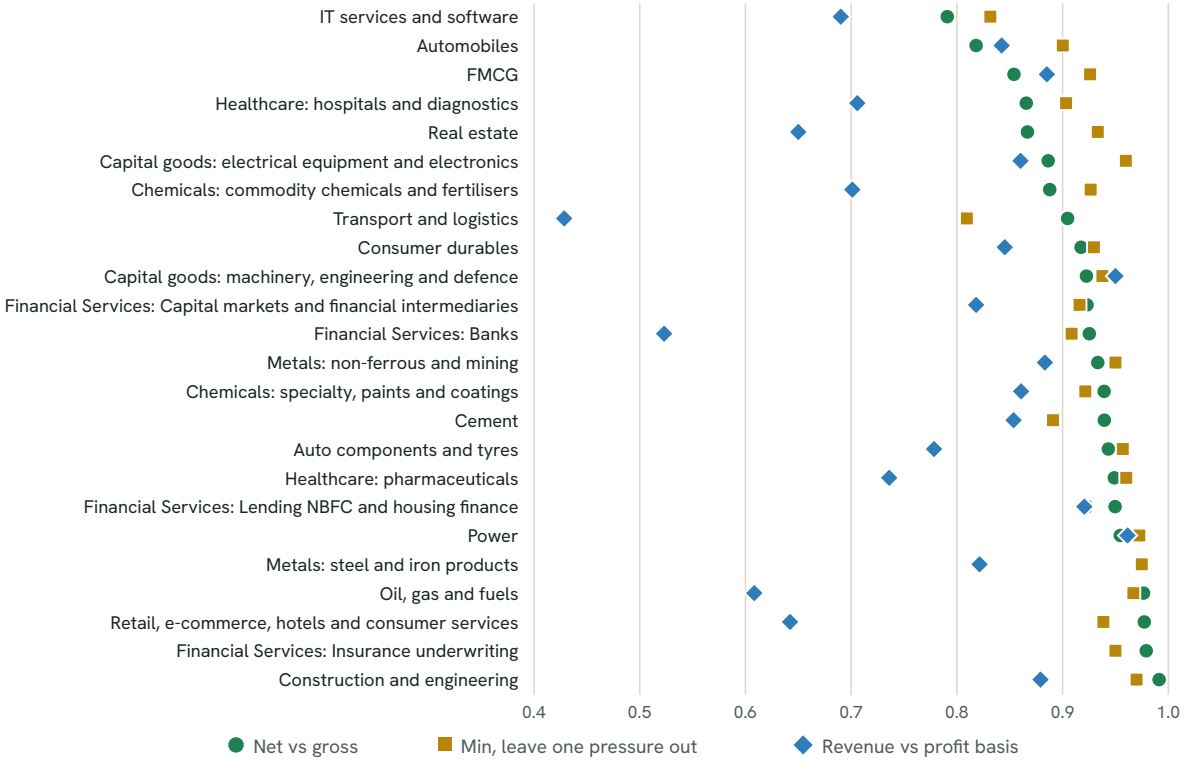


Figure 12.1. Rank stability by peer group. Spearman correlation between the published score and three alternatives; 1.0 means identical order.

Reading the table.

- **Offsets.** Net and gross ranks agree (median 0.93). Agreement is lowest in IT services and software (0.79) and automobiles (0.82), where renewable power and water recycling vary most.
- **Weights.** Dropping any one pressure changes ranks little (median of the group minima 0.93); the lowest is transport and logistics (0.81). No single pressure drives the ranking.
- **Denominator.** Revenue and profit ranks agree less (median 0.82), and least where margins vary most: transport and logistics (0.43) and banks (0.52). Insurance has no profit-basis ranks.
- **Untested choices.** We have not varied the thresholds of Section 6 (300×, 1.5×, 10×, 90%, the 20% tolerance), the minimum group size or the peer-group placements. Before publication we recommend a full uncertainty analysis that samples them together (Saisana et al. 2005).

13 Limitations

- **Pressure, as reported.** CNPI measures reported drivers of nature loss. It does not measure biodiversity or change in it.
- **Own operations.** We leave out supply chains, product use and financed impacts; for food, textiles and banks the operational footprint is a small part of the whole.
- **Self-reported data.** We take values as published. Auditors assure many of them under BRSR Core (SEBI 2023), but we do not audit them. More than half the ranked companies (221 of 409) rest on at least one value that needed reconciliation, a unit rule, a correction or a public-report fill.
- **Revenue as denominator** favours companies that sell high-value products. Cement and power have a physical alternative (Section 9); the other groups have none.
- **Mixed boundaries.** Standalone and consolidated filers sit in the same groups (312 and 96 among the ranked). With these data you cannot separate the boundary effect from business mix.
- **Undeclared Scope 2.** In practice most Scope 2 figures are location-based and use the previous year's grid factor (0.710 for FY2024-25; 0.675 for FY2025-26, CEA 2026). A within-filing test (Scope 2 ÷ purchased non-renewable electricity outside 0.5–1.0 tCO2/MWh) flags 42 companies, 34 of them ranked, that no hold-back rule catches.
- **Captive power** moves emissions between Scope 1 and Scope 2 according to who owns the plant, whatever it emits.
- **Unknown water location.** Company totals carry no basin, so a kilolitre in a stressed basin counts the same as one in an abundant basin.
- **"Recovered" waste** includes fly ash sent to cement and overburden used as backfill. A near-total offset may reward a reporting category as much as a practice; 62 companies have net waste of zero or less before the zero floor.
- **Small groups.** In groups of 8–10 companies, one changed value moves ranks more.
- **A narrow assumed-discharge rule.** Rule V7 scores water on withdrawal where consumption is 20% of withdrawal. We score other round ratios that may also be assumptions as reported, and the rule changes leaders in office-based groups.
- **Mixed Scope 2 conventions.** Rule V2 accepts a zero Scope 2 where the renewable share is at least 90%, which fits a market-based figure, while most filings are location-based in practice.
- **Naming cautions** flag 31 ranked companies whose business model, boundary or a reported value needs checking before anyone names them.
- **Equal weights** are a normative choice.
- **Percentile base.** We score each pressure over all companies with a usable value on it, including some whose rank we withhold (Section 7.2). A published score can depend on companies missing from the rank.
- **Untested thresholds.** See Section 12.
- **One year.** Version 1.2 is a snapshot and shows no change over time.

14 What a full corporate nature index would require

CNPI v1.2 uses what the BRSR provides, and a defensible corporate nature index needs more. Table 14.1 condenses the one-sheet comparison in `insights/pressures/DATA_NEEDS.md`, which maps each parameter against the TNFD recommendations (TNFD 2023, and the October 2026 core global indicators), GRI 101 (GRI 2024), IFRS S2 (ISSB 2023), ESRS and SBTN. Severity: *critical*, the BRSR does not ask and nothing usable exists; *high*, it asks in loose or voluntary terms and comparability fails; *medium*, the number exists but its metadata prevent clean comparison.

Table 14.1. Data needs by parameter.

Parameter	BRSR today	Observed in CNPI v1.2	Gap	The one disclosure ask
Climate	EI 7: Scope 1 and 2 totals; Scope 3 voluntary	473/500 with Scope 1+2; method stated in 16/465; Scope 3 resolvable in 141/465	Medium	State the Scope 2 method and grid-factor version, report location- and market-based separately, and state the GHG consolidation boundary
Energy	EI 1: electricity, fuel, other; renewable vs non-renewable	473/500; 38 values for 13 companies corrected by hand; captive generation not identifiable	Medium	Report grid-purchased and self-generated electricity on separate lines, naming the fuel already counted
Water quantity	EI 3–4: withdrawal by source, consumption, discharge (company level)	473/500; consumption above withdrawal in 26; assumed 80% discharge in 30	Medium	Compute consumption as withdrawal minus discharge (± storage); report seawater separately
Water location and stress	LI 1 (voluntary): area name only	No facility coordinates in the BRSR	High	Mandatory facility table with district, coordinates and the water-stress dataset named
Wastewater	EI 4: volume by destination and treatment	Pollutant loads not asked	High	Annual loads (COD, BOD, N, P) per discharge point
Waste	EI 9: eight categories; recovered and disposed by route	472/500; recovered > generated in 25; unscored in 4 groups (incl. real estate)	Medium	Named lines for fly ash, overburden, tailings and slag; recovery from stock reported separately
Air	EI 6: pollutants with "please specify unit"	Mass for 245/500; 111 companies' values not usable as mass	High	NOx, SOx and PM in tonnes per year
Land use	Number of plants and offices only	Company-wide land held for 13/500; site areas for 140 companies	Critical	Hectares owned, leased and under mining lease, plus area disturbed and restored in the year
Biodiversity and sensitive sites	EI 11 free-text table; LI 3 narrative	Coordinates for 10/326 sites; 18 affirmative sensitive-site statements	Critical	Coordinates (polygons for sites ≥ 10 ha) and area for every operating site
Value chain	LI 2 Scope 3 (voluntary); sourcing share in %	Commodity tonnes by origin not asked	Critical	Tonnes of high-impact natural commodities by type and origin, with share certified

The asks with the largest return concern **location and units**: site coordinates, water by facility, land area, air in tonnes and the Scope 2 method. Most companies already hold this information in environmental-clearance and consent documents, so the asks concern format more than new measurement.

15 Reproducibility

Scripts (all in `/home/claude/CLBI_work_2026-10-24/`, Python 3 with pandas, NumPy, SciPy and openpyxl):

Order	Script	Reads	Writes
1	`consolidate_gapfill.py`	gap-fill agent results	`inputs/gapfill_values.csv`, `research/gapfill_QA.csv`
2	`consolidate_energy.py`	energy agent results	`inputs/gapfill_energy.csv`, `research/energy_QA.csv`
3	`build_clbi.py` (with `v11_extensions.py`)	Codex snapshot and `inputs/`	`outputs_v12/`: scores, company data, data queries, robustness, workbook, `build_stats.json`
4	`land_evidence_v12.py`	`research/hermes/` and the scores file	`CLBI_v1_land_evidence.csv`, `CLBI_v1_land_sites.csv`
5	`exclusions_v12.py`	scores and data queries	`CLBI_v1_exclusions.csv`, `CLBI_v1_exclusion_summary.csv`
6	`make_dataset_v12.py`	`outputs_v12/`, `inputs/`	`deliverables_v12/CNPI_v1.2_Dataset.xlsx`

Rebuild command for step 3:

```
python3 build_clbi.py --codex /home/claude/codex --out outputs_v12 --inputs inputs
```

Parameters. The dictionary `P` at the top of `build_clbi.py` holds every judgement, and each build copies it into `build_stats.json` (Appendix B).

Input fingerprints. `build_stats.json` records the SHA-256 hash of the 10 frozen Codex inputs (`input_sha256`) and of every CSV in `inputs/` (`inputs_sha256`, 20 files at this build). A change to any input changes its hash. The fingerprints of this build begin: `normalized_companies.json` cc960e972362d4df; `Physical_Variable_Accounts_15500.csv` 8d94ff002faf1bd9; `financial_revenue_ledger_all500.csv` f4fb2656c6836e29; `peer_groups_v12.csv` 0c60579cb02a8c44; `gapfill_values.csv` 526c074cffab0195; `gapfill_energy.csv` 3cbc4eba65993355; `prices_v12.csv` 142cdfbab49c4e2a; `seawater_in_consumption.csv` 9fdbcd5211c0b0d9.

Spreadsheet check. The dataset workbook recomputes intensities, scores and ranks for all 500 companies with spreadsheet formulas, so you can check the arithmetic without Python.

Known code notes.

- `build_clbi.py` reads `inputs/query_resolutions.csv` (467 rows, 461 marked “confirmed”) and does not apply it: a confirmation does not release a held value in this build. Corrections reach the build through `gapfill_values.csv`.
- 41 gap-fill rows still show “page nan” in `source_field`: they record the source URL and lack the page.
- Version 1.2 sets the positive-action credit of v1.1 to zero (all `credit_*` parameters are 0); the code path remains.
- The `build_stats.json` field `sectors_ranked` (17) counts NSE sectors, after moves, with at least 8 members; it does not count peer groups.

Annex: Peer-group profiles

Each peer group has one profile, in order of size, and we computed every figure from `outputs_v12`. *Median intensity* is the median net value per ₹ crore of revenue among companies with a usable value. *Spread* is the highest usable positive climate intensity divided by the lowest. *Highest-ranked without caution* is the best-ranked company with no naming caution (Section 6, V13); we name it to illustrate the method, and the naming endorses nothing. Nature cost uses set B, as a median share of EBITDA (non-financial groups) or of revenue (financial groups), where we cost at least 5 companies.

Healthcare: pharmaceuticals

40 of 40 members ranked on climate, energy, water, pollution. Climate intensity spans 392× across members with a usable value.

Members / ranked / air-ranked	40 / 40 / 26
Drawn from NSE sectors	Healthcare (40)
Moved in from another NSE sector	0
Median intensity, climate	13.6 tCO2e/₹ cr (n = 40)
Median intensity, energy	79.9 GJ/₹ cr (n = 40)
Median intensity, water	77.6 kL/₹ cr (n = 40)
Median intensity, pollution	0.250 t/₹ cr (n = 40)
Highest-ranked without caution	GlaxoSmithkline Pharmaceuticals Ltd. (rank 4, score 89.1)
Ranked with a naming caution	5
Rank correlation: net vs gross / min leave-one-out	0.95 / 0.96
Illustrative nature cost, median	4.0% of EBITDA (n = 40)

Not ranked: none; every member is ranked.

Financial Services: Lending NBFC and housing finance

29 of 33 members ranked on climate, energy, water, pollution. Climate intensity spans 357× across members with a usable value.

Members / ranked / air-ranked	33 / 29 / 0
Drawn from NSE sectors	Financial Services (33)
Moved in from another NSE sector	0
Median intensity, climate	0.477 tCO2e/₹ cr (n = 32)
Median intensity, energy	2.37 GJ/₹ cr (n = 33)
Median intensity, water	4.00 kL/₹ cr (n = 31)
Median intensity, pollution	0.00045 t/₹ cr (n = 32)
Highest-ranked without caution	REC LIMITED (rank 1, score 93.1)
Ranked with a naming caution	0
Rank correlation: net vs gross / min leave-one-out	0.95 / 0.92
Illustrative nature cost, median	0.04% of revenue (n = 33)

Not ranked (4): Bajaj Finance Limited (held back); Bajaj Finserv Limited (held back); Home First Finance Company India Limited (not reported); Authum Investment & Infrastructure Limited (zero to confirm).

Auto components and tyres

27 of 29 members ranked on climate, energy, water, pollution. Climate intensity spans 224× across members with a usable value.

Members / ranked / air-ranked	29 / 27 / 15
Drawn from NSE sectors	Automobile and Auto Components (27); Capital Goods (1); Fast Moving Consumer Goods (1)
Moved in from another NSE sector	2
Median intensity, climate	18.4 tCO2e/₹ cr (n = 27)
Median intensity, energy	119 GJ/₹ cr (n = 27)
Median intensity, water	70.6 kL/₹ cr (n = 27)
Median intensity, pollution	0.178 t/₹ cr (n = 27)
Highest-ranked without caution	Tenneco Clean Air India Limited (rank 1, score 94.2)
Ranked with a naming caution	1
Rank correlation: net vs gross / min leave-one-out	0.94 / 0.96
Illustrative nature cost, median	9.0% of EBITDA (n = 27)

Not ranked (2): CIE Automotive India Limited (revenue mismatch); Schaeffler India Limited (revenue mismatch).

FMCG

27 of 28 members ranked on climate, energy, water, pollution. Climate intensity spans 954× across members with a usable value.

Members / ranked / air-ranked	28 / 27 / 22
Drawn from NSE sectors	Fast Moving Consumer Goods (28)
Moved in from another NSE sector	0
Median intensity, climate	6.24 tCO2e/₹ cr (n = 27)
Median intensity, energy	31.3 GJ/₹ cr (n = 27)
Median intensity, water	61.8 kL/₹ cr (n = 27)
Median intensity, pollution	0.032 t/₹ cr (n = 27)
Highest-ranked without caution	United Spirits Limited (rank 1, score 88.5)
Ranked with a naming caution	3
Rank correlation: net vs gross / min leave-one-out	0.85 / 0.93
Illustrative nature cost, median	4.4% of EBITDA (n = 27)

Not ranked (1): Varun Beverages Limited (revenue mismatch).

Capital goods: machinery, engineering and defence

25 of 31 members ranked on climate, energy, water, pollution. Climate intensity spans 320× across members with a usable value.

Members / ranked / air-ranked	31 / 25 / 11
Drawn from NSE sectors	Capital Goods (31)
Moved in from another NSE sector	0
Median intensity, climate	5.16 tCO2e/₹ cr (n = 27)
Median intensity, energy	38.6 GJ/₹ cr (n = 28)
Median intensity, water	50.3 kL/₹ cr (n = 29)
Median intensity, pollution	0.435 t/₹ cr (n = 30)
Highest-ranked without caution	Zen Technologies Ltd. (rank 1, score 82.2)
Ranked with a naming caution	2
Rank correlation: net vs gross / min leave-one-out	0.92 / 0.94
Illustrative nature cost, median	3.5% of EBITDA (n = 28)

Not ranked (6): BEML Ltd (held back); Ingersoll-Rand (India) Ltd. (held back); Thermax Ltd. (held back); SKF INDIA (INDUSTRIAL) LIMITED (not reported); Azad Engineering Limited (zero to confirm); KSB LIMITED (revenue mismatch).

Financial Services: Banks

25 of 27 members ranked on climate, energy, water, pollution. Climate intensity spans 34× across members with a usable value.

Members / ranked / air-ranked	27 / 25 / 0
Drawn from NSE sectors	Financial Services (27)
Moved in from another NSE sector	0
Median intensity, climate	1.17 tCO2e/₹ cr (n = 26)
Median intensity, energy	6.21 GJ/₹ cr (n = 26)
Median intensity, water	6.97 kL/₹ cr (n = 25)
Median intensity, pollution	0.0008 t/₹ cr (n = 26)
Highest-ranked without caution	ICICI Bank Ltd. (rank 1, score 84.8)
Ranked with a naming caution	0
Rank correlation: net vs gross / min leave-one-out	0.93 / 0.91
Illustrative nature cost, median	0.10% of revenue (n = 26)

Not ranked (2): Central Bank of India (held back); Jammu and Kashmir Bank Ltd. (held back).

Capital goods: electrical equipment and electronics

24 of 29 members ranked on climate, energy, water, pollution. Climate intensity spans 68× across members with a usable value.

Members / ranked / air-ranked	29 / 24 / 10
Drawn from NSE sectors	Capital Goods (23); Consumer Durables (3); Telecommunication (2); Information Technology (1)
Moved in from another NSE sector	6
Median intensity, climate	6.91 tCO2e/₹ cr (n = 24)
Median intensity, energy	37.7 GJ/₹ cr (n = 24)
Median intensity, water	15.8 kL/₹ cr (n = 24)
Median intensity, pollution	0.049 t/₹ cr (n = 24)
Highest-ranked without caution	NETWEB TECHNOLOGIES INDIA LIMITED (rank 1, score 94.6)
Ranked with a naming caution	1
Rank correlation: net vs gross / min leave-one-out	0.89 / 0.96
Illustrative nature cost, median	2.6% of EBITDA (n = 25)

Not ranked (5): Aditya Infotech Limited (not reported); Emmvee Photovoltaic Power Limited (not reported); ABB India Limited (revenue mismatch); Siemens Energy India Limited (revenue mismatch); Siemens Ltd. (revenue mismatch).

IT services and software

20 of 24 members ranked on climate, energy, water, pollution. Climate intensity spans 38× across members with a usable value.

Members / ranked / air-ranked	24 / 20 / 12
Drawn from NSE sectors	Information Technology (19); Financial Services (2); Services (2); Healthcare (1)
Moved in from another NSE sector	5
Median intensity, climate	0.849 tCO2e/₹ cr (n = 22)
Median intensity, energy	3.47 GJ/₹ cr (n = 20)
Median intensity, water	10.6 kL/₹ cr (n = 22)
Median intensity, pollution	0.0095 t/₹ cr (n = 21)
Highest-ranked without caution	Persistent Systems Limited (rank 1, score 75.6)
Ranked with a naming caution	0
Rank correlation: net vs gross / min leave-one-out	0.79 / 0.83
Illustrative nature cost, median	0.37% of EBITDA (n = 22)

Not ranked (4): KPIT Technologies Ltd (held back); Fractal Analytics Limited (not reported); Pine Labs Limited (not reported); Hexaware Technologies Limited (revenue mismatch).

Retail, e-commerce, hotels and consumer services

19 of 27 members ranked on climate, energy, water, pollution. Climate intensity spans 748× across members with a usable value.

Members / ranked / air-ranked	27 / 19 / 9
Drawn from NSE sectors	Consumer Services (24); Services (2); Consumer Durables (1)
Moved in from another NSE sector	3
Median intensity, climate	8.07 tCO2e/₹ cr (n = 21)
Median intensity, energy	48.2 GJ/₹ cr (n = 22)
Median intensity, water	34.0 kL/₹ cr (n = 22)
Median intensity, pollution	0.179 t/₹ cr (n = 20)
Highest-ranked without caution	Lenskart Solutions Limited (rank 3, score 77.3)
Ranked with a naming caution	2
Rank correlation: net vs gross / min leave-one-out	0.98 / 0.94
Illustrative nature cost, median	2.7% of EBITDA (n = 23)

Not ranked (8): Eternal Limited (held back); Metro Brands Limited (held back); ITC Hotels Limited (not reported); Physicswallah Limited (not reported); Urban Company Limited (not reported); BLS International Services Ltd (zero to confirm); Meesho Limited (zero to confirm); International Gemological Institute Limited (revenue mismatch).

Chemicals: commodity chemicals and fertilisers

17 of 18 members ranked on climate, energy, water, pollution. Climate intensity spans 152× across members with a usable value.

Members / ranked / air-ranked	18 / 17 / 13
Drawn from NSE sectors	Chemicals (15); Diversified (2); Capital Goods (1)
Moved in from another NSE sector	3
Median intensity, climate	61.0 tCO2e/₹ cr (n = 17)
Median intensity, energy	700 GJ/₹ cr (n = 18)
Median intensity, water	268 kL/₹ cr (n = 18)
Median intensity, pollution	0.414 t/₹ cr (n = 18)
Highest-ranked without caution	Supreme Petrochem Ltd. (rank 2, score 79.3)
Ranked with a naming caution	1
Rank correlation: net vs gross / min leave-one-out	0.89 / 0.93
Illustrative nature cost, median	50.8% of EBITDA (n = 17)

Not ranked (1): Himadri Speciality Chemical Ltd. (zero to confirm).

Chemicals: specialty, paints and coatings

15 of 17 members ranked on climate, energy, water, pollution. Climate intensity spans 93× across members with a usable value.

Members / ranked / air-ranked	17 / 15 / 13
Drawn from NSE sectors	Chemicals (11); Consumer Durables (4); Diversified (1); Oil Gas & Consumable Fuels (1)
Moved in from another NSE sector	6
Median intensity, climate	38.2 tCO2e/₹ cr (n = 16)
Median intensity, energy	344 GJ/₹ cr (n = 16)
Median intensity, water	70.9 kL/₹ cr (n = 15)
Median intensity, pollution	0.357 t/₹ cr (n = 15)
Highest-ranked without caution	JSW Dulux Limited (rank 1, score 85.7)
Ranked with a naming caution	1
Rank correlation: net vs gross / min leave-one-out	0.94 / 0.92
Illustrative nature cost, median	11.1% of EBITDA (n = 16)

Not ranked (2): PRIVI SPECIALITY CHEMICALS LIMITED (held back); Castrol India Ltd (revenue mismatch).

Metals: steel and iron products

15 of 17 members ranked on climate, energy, water, pollution. Climate intensity spans 309× across members with a usable value.

Members / ranked / air-ranked	17 / 15 / 9
Drawn from NSE sectors	Metals & Mining (8); Capital Goods (7); Automobile and Auto Components (1); Power (1)
Moved in from another NSE sector	9
Median intensity, climate	478 tCO2e/₹ cr (n = 16)
Median intensity, energy	4,674 GJ/₹ cr (n = 16)
Median intensity, water	615 kL/₹ cr (n = 17)
Median intensity, pollution	2.86 t/₹ cr (n = 17)
Highest-ranked without caution	Jindal Stainless Limited (rank 3, score 81.4)
Ranked with a naming caution	4
Rank correlation: net vs gross / min leave-one-out	0.97 / 0.97
Illustrative nature cost, median	165.0% of EBITDA (n = 16)

Not ranked (2): GALLANTT ISPAT LIMITED (held back); Shyam Metalics and Energy Limited (held back).

Power

13 of 17 members ranked on climate, energy, water. Climate intensity spans 1,202× across members with a usable value.

Members / ranked / air-ranked	17 / 13 / 8
Drawn from NSE sectors	Power (17)
Moved in from another NSE sector	0
Median intensity, climate	325 tCO2e/₹ cr (n = 14)
Median intensity, energy	5,693 GJ/₹ cr (n = 15)
Median intensity, water	2,401 kL/₹ cr (n = 14)
Median intensity, pollution (shown, not scored)	0.036 t/₹ cr (n = 10)
Highest-ranked without caution	NHPC Limited (rank 2, score 89.9)
Ranked with a naming caution	5
Rank correlation: net vs gross / min leave-one-out	0.95 / 0.97
Illustrative nature cost, median	159.9% of EBITDA (n = 15)

Not ranked (4): Adani Energy Solutions Limited (held back); Vedanta Power Limited (not reported); Reliance Power Limited (zero to confirm); Clean Max Enviro Energy Solutions Limited (revenue mismatch).

Oil, gas and fuels

13 of 15 members ranked on climate, energy, water, pollution. Climate intensity spans 62× across members with a usable value.

Members / ranked / air-ranked	15 / 13 / 8
Drawn from NSE sectors	Oil Gas & Consumable Fuels (15)
Moved in from another NSE sector	0
Median intensity, climate	25.1 tCO2e/₹ cr (n = 14)
Median intensity, energy	294 GJ/₹ cr (n = 14)
Median intensity, water	85.8 kL/₹ cr (n = 14)
Median intensity, pollution	0.078 t/₹ cr (n = 13)
Highest-ranked without caution	Aegis Logistics Ltd. (rank 1, score 92.0)
Ranked with a naming caution	1
Rank correlation: net vs gross / min leave-one-out	0.98 / 0.97
Illustrative nature cost, median	15.5% of EBITDA (n = 13)

Not ranked (2): Indraprashtha Gas Ltd. (held back); Vedanta Oil and Gas Limited (not reported).

Financial Services: Capital markets and financial intermediaries

12 of 18 members ranked on climate, energy, water, pollution. Climate intensity spans 88× across members with a usable value.

Members / ranked / air-ranked	18 / 12 / 0
Drawn from NSE sectors	Financial Services (18)
Moved in from another NSE sector	0
Median intensity, climate	0.899 tCO2e/₹ cr (n = 14)
Median intensity, energy	4.41 GJ/₹ cr (n = 16)
Median intensity, water	4.27 kL/₹ cr (n = 17)
Median intensity, pollution	0.00088 t/₹ cr (n = 15)
Highest-ranked without caution	360 ONE WAM LIMITED (rank 1, score 76.7)
Ranked with a naming caution	0
Rank correlation: net vs gross / min leave-one-out	0.92 / 0.92
Illustrative nature cost, median	0.07% of revenue (n = 15)

Not ranked (6): Aditya Birla Sun Life AMC Limited (held back); Choice International Ltd (held back); Anand Rathi Wealth Limited (zero to confirm); ICICI Prudential Asset Management Company Limited (zero to confirm); Nuvama Wealth Management Limited (zero to confirm); CRISIL Ltd. (revenue mismatch).

Automobiles

12 of 14 members ranked on climate, energy, water, pollution. Climate intensity spans 28× across members with a usable value.

Members / ranked / air-ranked	14 / 12 / 12
Drawn from NSE sectors	Automobile and Auto Components (11); Capital Goods (3)
Moved in from another NSE sector	3
Median intensity, climate	2.16 tCO2e/₹ cr (n = 13)
Median intensity, energy	16.0 GJ/₹ cr (n = 13)
Median intensity, water	19.2 kL/₹ cr (n = 13)
Median intensity, pollution	0.084 t/₹ cr (n = 12)
Highest-ranked without caution	TATA MOTORS PASSENGER VEHICLES LIMITED (rank 1, score 86.7)
Ranked with a naming caution	0
Rank correlation: net vs gross / min leave-one-out	0.82 / 0.90
Illustrative nature cost, median	1.2% of EBITDA (n = 11)

Not ranked (2): TVS Motor Company Ltd. (held back); Ather Energy Limited (not reported).

Construction and engineering

12 of 13 members ranked on climate, energy, water. Climate intensity spans 59× across members with a usable value.

Members / ranked / air-ranked	13 / 12 / 0
Drawn from NSE sectors	Construction (11); Telecommunication (1); Services (1)
Moved in from another NSE sector	2
Median intensity, climate	5.17 tCO2e/₹ cr (n = 12)
Median intensity, energy	49.8 GJ/₹ cr (n = 12)
Median intensity, water	72.3 kL/₹ cr (n = 12)
Median intensity, pollution (shown, not scored)	0.943 t/₹ cr (n = 8)
Highest-ranked without caution	IRCON International Ltd (rank 1, score 93.9)
Ranked with a naming caution	0
Rank correlation: net vs gross / min leave-one-out	0.99 / 0.97
Illustrative nature cost, median	4.7% of EBITDA (n = 12)

Not ranked (1): Central Mine Planning & Design Institute Limited (not reported).

Cement

10 of 10 members ranked on climate, energy, water, pollution. Climate intensity spans 3× across members with a usable value.

Members / ranked / air-ranked	10 / 10 / 10
Drawn from NSE sectors	Construction Materials (10)
Moved in from another NSE sector	0
Median intensity, climate	1,020 tCO2e/₹ cr (n = 10)
Median intensity, energy	4,258 GJ/₹ cr (n = 10)
Median intensity, water	243 kL/₹ cr (n = 10)
Median intensity, pollution	7.7e-05 t/₹ cr (n = 10)
Highest-ranked without caution	ACC Ltd (rank 1, score 86.1)
Ranked with a naming caution	0
Rank correlation: net vs gross / min leave-one-out	0.94 / 0.89
Illustrative nature cost, median	316.0% of EBITDA (n = 10)

Not ranked: none; every member is ranked.

Consumer durables

10 of 10 members ranked on climate, energy, water, pollution. Climate intensity spans 71× across members with a usable value.

Members / ranked / air-ranked	10 / 10 / 0
Drawn from NSE sectors	Consumer Durables (9); Fast Moving Consumer Goods (1)
Moved in from another NSE sector	1
Median intensity, climate	2.28 tCO2e/₹ cr (n = 10)
Median intensity, energy	13.3 GJ/₹ cr (n = 10)
Median intensity, water	13.3 kL/₹ cr (n = 10)
Median intensity, pollution	0.040 t/₹ cr (n = 10)
Highest-ranked without caution	Crompton Greaves Consumer Electricals Limited (rank 3, score 66.7)
Ranked with a naming caution	4
Rank correlation: net vs gross / min leave-one-out	0.92 / 0.93
Illustrative nature cost, median	2.3% of EBITDA (n = 10)

Not ranked: none; every member is ranked.

Metals: non-ferrous and mining

9 of 13 members ranked on climate, energy, water. Climate intensity spans 303× across members with a usable value.

Members / ranked / air-ranked	13 / 9 / 0
Drawn from NSE sectors	Metals & Mining (11); Oil Gas & Consumable Fuels (2)
Moved in from another NSE sector	2
Median intensity, climate	110 tCO2e/₹ cr (n = 10)
Median intensity, energy	1,074 GJ/₹ cr (n = 11)
Median intensity, water	633 kL/₹ cr (n = 10)
Median intensity, pollution (shown, not scored)	335 t/₹ cr (n = 9)
Highest-ranked without caution	NMDC Ltd (rank 1, score 89.3)
Ranked with a naming caution	0
Rank correlation: net vs gross / min leave-one-out	0.93 / 0.95
Illustrative nature cost, median	30.4% of EBITDA (n = 10)

Not ranked (4): Coal India Limited (held back); Jain Resource Recycling Limited (held back); BHARAT COKING COAL LIMITED (not reported); Vedanta Aluminium Metal Limited (revenue mismatch).

Financial Services: Insurance underwriting

9 of 12 members ranked on climate, energy, water, pollution. Climate intensity spans 22× across members with a usable value.

Members / ranked / air-ranked	12 / 9 / 0
Drawn from NSE sectors	Financial Services (12)
Moved in from another NSE sector	0
Median intensity, climate	0.195 tCO2e/₹ cr (n = 11)
Median intensity, energy	0.925 GJ/₹ cr (n = 12)
Median intensity, water	3.63 kL/₹ cr (n = 12)
Median intensity, pollution	0.00058 t/₹ cr (n = 10)
Highest-ranked without caution	ICICI Prudential Life Insurance Company Limited (rank 1, score 78.8)
Ranked with a naming caution	0
Rank correlation: net vs gross / min leave-one-out	0.98 / 0.95
Illustrative nature cost, median	0.03% of revenue (n = 11)

Not ranked (3): Canara HSBC Life Insurance Company Limited (held back); General Insurance Corporation of India (zero to confirm); The New India Assurance Company Limited (zero to confirm).

Real estate

9 of 12 members ranked on climate, energy, water. Climate intensity spans 25× across members with a usable value.

Members / ranked / air-ranked	12 / 9 / 0
Drawn from NSE sectors	Realty (11); Media Entertainment & Publication (1)
Moved in from another NSE sector	1
Median intensity, climate	5.93 tCO2e/₹ cr (n = 9)
Median intensity, energy	30.7 GJ/₹ cr (n = 11)
Median intensity, water	101 kL/₹ cr (n = 11)
Median intensity, pollution (shown, not scored)	0.513 t/₹ cr (n = 10)
Highest-ranked without caution	Anant Raj Limited (rank 1, score 85.8)
Ranked with a naming caution	0
Rank correlation: net vs gross / min leave-one-out	0.87 / 0.93
Illustrative nature cost, median	2.2% of EBITDA (n = 10)

Not ranked (3): ADITYA BIRLA REAL ESTATE LIMITED (held back); Prime Focus Limited (held back); Lodha Developers Limited (zero to confirm).

Healthcare: hospitals and diagnostics

9 of 10 members ranked on climate, energy, water, pollution. Climate intensity spans 3× across members with a usable value.

Members / ranked / air-ranked	10 / 9 / 0
Drawn from NSE sectors	Healthcare (10)
Moved in from another NSE sector	0
Median intensity, climate	8.84 tCO2e/₹ cr (n = 9)
Median intensity, energy	44.0 GJ/₹ cr (n = 9)
Median intensity, water	136 kL/₹ cr (n = 9)
Median intensity, pollution	0.313 t/₹ cr (n = 9)
Highest-ranked without caution	Dr. Lal Pathlabs Limited (rank 1, score 84.4)
Ranked with a naming caution	1
Rank correlation: net vs gross / min leave-one-out	0.87 / 0.90
Illustrative nature cost, median	4.3% of EBITDA (n = 9)

Not ranked (1): Dr. Agarwal's Health Care Limited (not reported).

Transport and logistics

8 of 10 members ranked on climate, energy, water, pollution. Climate intensity spans 86× across members with a usable value.

Members / ranked / air-ranked	10 / 8 / 0
Drawn from NSE sectors	Services (9); Oil Gas & Consumable Fuels (1)
Moved in from another NSE sector	1
Median intensity, climate	23.3 tCO2e/₹ cr (n = 9)
Median intensity, energy	323 GJ/₹ cr (n = 9)
Median intensity, water	39.3 kL/₹ cr (n = 9)
Median intensity, pollution	0.025 t/₹ cr (n = 8)
Highest-ranked without caution	GMR Airports Limited (rank 1, score 78.1)
Ranked with a naming caution	0
Rank correlation: net vs gross / min leave-one-out	0.90 / 0.81
Illustrative nature cost, median	14.6% of EBITDA (n = 9)

Not ranked (2): Container Corporation Of India Ltd. (held back); AEGIS VOPAK TERMINALS LIMITED (not reported).

Telecom and media

Not ranked: 8 members, fewer than 8 of them with complete, clean data. Climate intensity spans 667× across members with a usable value.

Members / ranked / air-ranked	8 / 0 / 0
Drawn from NSE sectors	Telecommunication (5); Consumer Services (2); Media Entertainment & Publication (1)
Moved in from another NSE sector	3
Median intensity, climate	2.46 tCO2e/₹ cr (n = 7)
Median intensity, energy	14.0 GJ/₹ cr (n = 8)
Median intensity, water	4.80 kL/₹ cr (n = 8)
Median intensity, pollution	0.00065 t/₹ cr (n = 8)
Illustrative nature cost, median	0.52% of EBITDA (n = 8)

Not ranked (8): Bharti Airtel Ltd. (group too small); Bharti Hexacom Limited (group too small); INDUS TOWERS LIMITED (group too small); Info Edge(India) Ltd. (group too small); SUN TV NETWORK LIMITED (group too small); Tata Communications Limited (group too small); VODAFONE IDEA LIMITED (group too small); IndiaMART InterMESH Limited (zero to confirm).

Textiles and apparel

Not ranked: 7 members, below the minimum of 8. Climate intensity spans 55× across members with a usable value.

Members / ranked / air-ranked	7 / 0 / 0
Drawn from NSE sectors	Textiles (5); Construction Materials (1); Chemicals (1)
Moved in from another NSE sector	2
Median intensity, climate	105 tCO2e/₹ cr (n = 7)
Median intensity, energy	648 GJ/₹ cr (n = 7)
Median intensity, water	273 kL/₹ cr (n = 7)
Median intensity, pollution	1.53 t/₹ cr (n = 7)
Illustrative nature cost, median	42.4% of EBITDA (n = 7)

Not ranked (7): Grasim Industries Ltd (group too small); K.P.R. Mill Ltd. (group too small); Page Industries Ltd. (group too small); Swan Corp Limited (group too small); Trident Ltd (group too small); Vardhman Textiles Limted (group too small); WELSPUN LIVING LIMITED (group too small).

Financial Services: Investment and holding companies

Not ranked: 6 members, below the minimum of 8.

Members / ranked / air-ranked	6 / 0 / 0
Drawn from NSE sectors	Financial Services (6)
Moved in from another NSE sector	0
Median intensity, climate	0.048 tCO2e/₹ cr (n = 1)
Median intensity, energy	0.492 GJ/₹ cr (n = 3)
Median intensity, water	1.07 kL/₹ cr (n = 2)
Median intensity, pollution	0.0029 t/₹ cr (n = 2)

Not ranked (6): Bajaj Holdings & Investment Limited (group too small); JSW Holdings Limited (group too small); Jio Financial Services Limited (group too small); Maharashtra Scooters Ltd. (group too small); TVS Holdings Limited (group too small); Tata Investment Corporation Ltd. (group too small).

Building products

Not ranked: 4 members, below the minimum of 8. Climate intensity spans 7× across members with a usable value.

Members / ranked / air-ranked	4 / 0 / 0
Drawn from NSE sectors	Capital Goods (2); Consumer Durables (2)
Moved in from another NSE sector	4
Median intensity, climate	26.4 tCO2e/₹ cr (n = 4)
Median intensity, energy	154 GJ/₹ cr (n = 4)
Median intensity, water	120 kL/₹ cr (n = 4)
Median intensity, pollution	0.923 t/₹ cr (n = 4)

Not ranked (4): ASTRAL LIMITED (group too small); Century Plyboards (India) Ltd. (group too small); Kajaria Ceramics Ltd (group too small); Supreme Industries Ltd. (group too small).

Diversified conglomerates

Not ranked: 1 member, below the minimum of 8.

Members / ranked / air-ranked	1 / 0 / 0
Drawn from NSE sectors	Metals & Mining (1)
Moved in from another NSE sector	1
Median intensity, climate	11.3 tCO2e/₹ cr (n = 1)
Median intensity, energy	92.0 GJ/₹ cr (n = 1)
Median intensity, water	81.3 kL/₹ cr (n = 1)
Median intensity, pollution	1.83 t/₹ cr (n = 1)

Not ranked (1): Adani Enterprises Ltd. (group too small).

References

Alkemade, R., van Oorschot, M., Miles, L., Nellemann, C., Bakkenes, M. and ten Brink, B. (2009). GLOBIO3: a framework to investigate options for reducing global terrestrial biodiversity loss. *Ecosystems* 12(3), 374–390. https://doi.org/10.1007/s10021-009-9229-5

CEA, Central Electricity Authority (2025). *CO2 Baseline Database for the Indian Power Sector, User Guide, Version 21.0* (FY2024-25 weighted-average emission factor 0.710 tCO2/MWh). New Delhi: Ministry of Power. https://cea.nic.in/wp-content/uploads/baseline/2025/12/User_Guide_V_21.0.pdf

CEA, Central Electricity Authority (2026). *CO2 Baseline Database for the Indian Power Sector, User Guide, Version 22.0* (FY2025-26 factor 0.675 tCO2/MWh). https://cea.nic.in/wp-content/uploads/baseline/2026/09/User_Guide__Version_22.0.pdf

CE Delft (de Vries, J. et al.) (2024). *Environmental Prices Handbook 2024, EU27 version 1.1*. Delft: CE Delft. https://cedelft.eu/wp-content/uploads/sites/2/2024/12/CE_Delft_230107_Environmental-Prices-Handbook-2024-EU-version_def_V1.1.pdf

GRI, Global Reporting Initiative (2024). *GRI 101: Biodiversity 2024* (effective for reports published on or after 1 January 2026). Amsterdam: GSSB. https://www.globalreporting.org/pdf.ashx?id=24534

IFVI and VBA (2026). *Environmental Methodology 2: Water Consumption*. International Foundation for Valuing Impacts and Value Balancing Alliance. https://ifvi.org/wp-content/uploads/2026/03/IFVI_VBA_Environmental_Methodology-2_Water-Consumption_FIN_04.pdf

IFVI (2024). *Global Value Factor Database*, version 1 (exposure draft, 15 October 2024), interim air and waste methodologies. International Foundation for Valuing Impacts. https://ifvi.org/methodology/environmental-topic-methodology/interim-methodologies/ (read via a CC-BY mirror, as documented in `PRICES_V12.md`)

IPBES (2019). *Summary for policymakers of the global assessment report on biodiversity and ecosystem services* (Díaz, S., Settele, J., Brondízio, E.S., Ngo, H.T. et al., eds). Bonn: IPBES secretariat. https://www.ipbes.net/sites/default/files/2020-02/ipbes_global_assessment_report_summary_for_policymakers_en.pdf

ISSB, International Sustainability Standards Board (2023). *IFRS S2 Climate-related Disclosures*. London: IFRS Foundation. https://www.ifrs.org/issued-standards/ifrs-sustainability-standards-navigator/ifrs-s2-climate-related-disclosures

IUCN (2016). *A Global Standard for the Identification of Key Biodiversity Areas, Version 1.0*. Gland: IUCN. https://portals.iucn.org/library/sites/library/files/documents/2016-048.pdf

Kumar, S. (2006). Analysing industrial water demand in India: an input distance function approach. *Water Policy* 8(1), 15–29. https://doi.org/10.2166/wp.2006.0002

Kuzma, S., Bierkens, M.F.P., Lakshman, S., Luo, T., Saccoccia, L., Sutanudjaja, E.H. and Van Beek, R. (2023). *Aqueduct 4.0: Updated decision-relevant global water risk indicators*. Technical Note. Washington, DC: World Resources Institute. https://doi.org/10.46830/writn.23.00061

MOSPI, Central Statistical Organisation (2008). *National Industrial Classification (All Economic Activities) 2008*. New Delhi: Ministry of Statistics and Programme Implementation. https://www.mospi.gov.in/sites/default/files/main_menu/national_industrial_classification/nic_2008_17apr09.pdf

OECD and JRC (Nardo, M., Saisana, M., Saltelli, A., Tarantola, S., Hoffmann, A. and Giovannini, E.) (2008). *Handbook on Constructing Composite Indicators: Methodology and User Guide*. Paris: OECD Publishing. https://doi.org/10.1787/9789264043466-en

Rennert, K., Errickson, F., Prest, B.C., Rennels, L., Newell, R.G., Pizer, W. et al. (2022). Comprehensive evidence implies a higher social cost of CO2. *Nature* 610, 687–692. https://doi.org/10.1038/s41586-022-05224-9

Ricke, K., Drouet, L., Caldeira, K. and Tavoni, M. (2018). Country-level social cost of carbon. *Nature Climate Change* 8, 895–900. https://doi.org/10.1038/s41558-018-0282-y

Saisana, M., Saltelli, A. and Tarantola, S. (2005). Uncertainty and sensitivity analysis techniques as tools for the quality assessment of composite indicators. *Journal of the Royal Statistical Society: Series A* 168(2), 307–323. https://doi.org/10.1111/j.1467-985X.2005.00350.x

Schipper, A.M., Hilbers, J.P., Meijer, J.R., Antão, L.H., Benítez-López, A., de Jonge, M.M.J. et al. (2020). Projecting terrestrial biodiversity intactness with GLOBIO 4. *Global Change Biology* 26(2), 760–771. https://doi.org/10.1111/gcb.14848

SEBI, Securities and Exchange Board of India (2021). *Business responsibility and sustainability reporting by listed entities*. Circular SEBI/HO/CFD/CMD-2/P/CIR/2021/562, 10 May 2021. https://www.sebi.gov.in/legal/circulars/may-2021/business-responsibility-and-sustainability-reporting-by-listed-entities_50096.html

SEBI (2023). *BRSR Core: Framework for assurance and ESG disclosures for value chain*. Circular SEBI/HO/CFD/CFD-SEC-2/P/CIR/2023/122, 12 July 2023. https://www.sebi.gov.in/legal/circulars/jul-2023/brsr-core-framework-for-assurance-and-esg-disclosures-for-value-chain_73854.html

Sotos, M. (2015). *GHG Protocol Scope 2 Guidance: An amendment to the GHG Protocol Corporate Standard*. Washington, DC: World Resources Institute. https://ghgprotocol.org/scope-2-guidance

Spearman, C. (1904). The proof and measurement of association between two things. *American Journal of Psychology* 15(1), 72–101. https://doi.org/10.2307/1412159

Thaller, A. et al. (2026). Country social cost of carbon estimates and their application to assess the natural land sink. Kiel Working Paper 2316. Kiel Institute for the World Economy (working paper, as cited in `PRICES_V12.md`).

TNFD, Taskforce on Nature-related Financial Disclosures (2023). *Recommendations of the Taskforce on Nature-related Financial Disclosures*, September 2023. https://tnfd.global/wp-content/uploads/2023/08/Recommendations_of_the_Taskforce_on_Nature-related_Financial_Disclosures_September_2023.pdf

TNFD (2026). *Recommendations of the Taskforce on Nature-related Financial Disclosures*, October 2026, Annex 1: core global disclosure metrics. https://tnfd.global/wp-content/uploads/2026/10/Recommendations-of-the-TNFD_October-2026_DIGITAL.pdf

Tol, R.S.J. (2019). A social cost of carbon for (almost) every country. *Energy Economics* 83, 555–566. https://doi.org/10.1016/j.eneco.2019.07.006

Tripathi, Gupta, Krishna and Gandhi (2026). *Business responsibility and sustainability in India: a comprehensive analysis of BRSR disclosures of India's top 1,000 listed companies for FY 2024–25*. Monograph. Bengaluru: Indian Institute of Management Bangalore. https://research.iimb.ac.in/cgi/viewcontent.cgi?article=1149&context=monographs

UNEP-WCMC and IUCN (2026). *Protected Planet: The World Database on Protected and Conserved Areas (WDPCA)*, October 2026. Cambridge, UK: UNEP-WCMC and IUCN. https://www.protectedplanet.net

US EPA (2023). *Report on the Social Cost of Greenhouse Gases: Estimates Incorporating Recent Scientific Advances*. Washington, DC: US Environmental Protection Agency. https://www.epa.gov/system/files/documents/2023-12/epa_scghg_2023_report_final.pdf

WRI and WBCSD (2004). *The Greenhouse Gas Protocol: A Corporate Accounting and Reporting Standard*, revised edition. Washington, DC and Geneva. https://ghgprotocol.org/sites/default/files/standards/ghg-protocol-revised.pdf

Prices in Table 11.1 also rest on further sources listed and verified in `insights/prices/PRICES_V12.md` (Bierkens et al. 2019; Huijbregts et al. 2017; Rabl et al. 2008; Dijkgraaf and Vollebergh 2004; COWI 2000; Nahman 2011; Jardosh and Kathuria 2025).

Appendix A. Glossary

Term	Meaning in this paper
AMFI	Association of Mutual Funds in India; publishes the six-monthly market-capitalisation list that defines the universe
BRSR	Business Responsibility and Sustainability Report, filed annually by India's top 1,000 listed companies under SEBI (2021)
BRSR Core	The subset of BRSR indicators subject to assurance (SEBI 2023)
Boundary	Whether a BRSR covers the listed company alone (standalone) or its group (consolidated)
Confidence tier	How a value reached the dataset (Table 3.2), recorded as `source_status`
Gross figure	The pressure before offsets: total energy, water withdrawn, waste generated
Held back	A reported value not scored until the company confirms it (Section 6)
Intensity	A net pressure divided by revenue from operations in ₹ crore
kL	Kilolitre, equal to one cubic metre
MSA	Mean species abundance: mean abundance of original species relative to an undisturbed state (GLOBIO)
Naming caution	A flag that a ranked company should not be singled out by name until a value or its business model is checked
Net figure	The pressure after the matching offset: non-renewable energy, water consumed, waste not recovered
NIC 2008	India's National Industrial Classification, used in BRSR product lines
Peer group	The set of companies with the same kind of main business within which ranks are computed
Percentile score	100 × (k − position) ÷ (k − 1), Section 7.2
Pressure	A reported quantity of a direct driver of nature loss
Scope 1, 2, 3	Direct emissions; emissions from purchased electricity, heat and steam; all other value-chain emissions (GHG Protocol)
SCC	Social cost of carbon: the monetised damage of one more tonne of CO2
tCO2e	Tonnes of carbon-dioxide equivalent
XBRL	The machine-readable filing format in which the BRSR is filed with NSE
₹ crore	Ten million rupees

Appendix B. Parameter table

All parameters are in `P` in `build_clbi.py` and in `build_stats.json` → `params`.

Parameter	Value	Used in	Rationale
`min_sector_size`	8	Section 5	Smallest group in which a within-group position is informative; lowered from 10 in v1.1
`min_element_reporters`	6	Section 7.2	A pressure is scored only if enough peers report it
`min_ranked_to_publish`	8	Section 7.4	Ranks published only with 8 complete, clean companies
`core_elements`	climate, energy, water, waste	Section 4	The four scored pressures; all required
`supplementary_elements`	air	Section 9	Ranked separately
`pollution_not_comparable`	Power; Metals: non-ferrous and mining; Construction and engineering; Real estate	V9	Largest waste streams reported inconsistently
`scope1_fuel_max_t_per_gj`	0.12 tCO2/GJ	V12	Fuel combustion emits at most about 0.1 t per GJ
`process_emission_groups`	Cement; Metals: steel and iron products; Metals: non-ferrous and mining; Chemicals: commodity chemicals and fertilisers; Oil, gas and fuels	V12	Groups where process emissions are expected, exempt from V12
`exclude_reported_zero`	climate, air	V1	Zero totals treated as “to confirm”
`extreme_ratio`	300	V3, V5, V13	Separates unit or boundary errors from genuine variation
`scope2_grid_ef`	0.71 tCO2/MWh	V4, V5	CEA FY2024-25 weighted-average grid factor
`scope2_tariff_inr_per_kwh`	₹6.0	V4	Conservative grid tariff for the bill test
`scope2_energy_ratio_max`	1.5	V5	Margin for grid-factor differences
`scope2_zero_re_min`	0.90	V2	Renewable share consistent with a market-based zero
`assumed_consumption_share`	0.20 (tolerance 0.0005)	V7	Detects an assumed 80% discharge
(in code) water cap band	1.05× to 10× withdrawal	V6	Consumption above withdrawal counts recycled water
(in code) water hold	> 10× withdrawal	V6	Likely unit error
(in code) waste hold	recovered > 1.5× generated	V8	Inconsistent table or stock movement
(in code) energy hold	non-renewable > 1.01× total	V10	Inconsistent table
(in code) naming-caution band	100× to 300× from peer median	V13	Unusual but inside the hold-back line
`thin_profit_margin`	0.02	Section 9	Profit-basis ratio unstable below a 2% margin
(in code) EBITDA thin margin	EBITDA < 5% of revenue	Section 11	% of EBITDA flagged as unstable
`credit_*`	0	–	v1.1 capped credit retired; offsets now act on the same pressure
`credit_a_min_ha`	10 ha	–	Retained for the retired credit; not used
Ranking rounding	6 decimal places	Section 7.4	Prevents floating-point noise from splitting ties

Appendix C. Peer groups: companies, ranked and not ranked

Counts from `CLBI_v1_exclusion_summary.csv`. The columns after "Not ranked" give the category of each company we do not rank (Section 8).

Peer group	Companies	Ranked	Not ranked	Held back	Zero to confirm	Not reported	Revenue mismatch	Group too small
Healthcare: pharmaceuticals	40	40	0	0	0	0	0	0
Financial Services: Lending NBFC and housing finance	33	29	4	2	1	1	0	0
Auto components and tyres	29	27	2	0	0	0	2	0
FMCG	28	27	1	0	0	0	1	0
Capital goods: machinery, engineering and defence	31	25	6	3	1	1	1	0
Financial Services: Banks	27	25	2	2	0	0	0	0
Capital goods: electrical equipment and electronics	29	24	5	0	0	2	3	0
IT services and software	24	20	4	1	0	2	1	0
Retail, e-commerce, hotels and consumer services	27	19	8	2	2	3	1	0
Chemicals: commodity chemicals and fertilisers	18	17	1	0	1	0	0	0
Chemicals: specialty, paints and coatings	17	15	2	1	0	0	1	0
Metals: steel and iron products	17	15	2	2	0	0	0	0
Power	17	13	4	1	1	1	1	0
Oil, gas and fuels	15	13	2	1	0	1	0	0
Financial Services: Capital markets and financial intermediaries	18	12	6	2	3	0	1	0
Automobiles	14	12	2	1	0	1	0	0
Construction and engineering	13	12	1	0	0	1	0	0
Cement	10	10	0	0	0	0	0	0
Consumer durables	10	10	0	0	0	0	0	0
Metals: non-ferrous and mining	13	9	4	2	0	1	1	0
Financial Services: Insurance underwriting	12	9	3	1	2	0	0	0
Real estate	12	9	3	2	1	0	0	0
Healthcare: hospitals and diagnostics	10	9	1	0	0	1	0	0
Transport and logistics	10	8	2	1	0	1	0	0
Telecom and media	8	0	8	0	1	0	0	7
Textiles and apparel	7	0	7	0	0	0	0	7
Financial Services: Investment and holding companies	6	0	6	0	0	0	0	6
Building products	4	0	4	0	0	0	0	4
Diversified conglomerates	1	0	1	0	0	0	0	1
Total	**500**	**409**	**91**	**24**	**13**	**16**	**13**	**25**

Appendix D. Air-ranked and profit-ranked counts

Table 5.1 gives the counts in its “Air-ranked” and “Profit-ranked” columns: air ranks in 14 groups (178 companies) and profit ranks in 22 groups (378 companies).

Appendix E. How we computed the less obvious numbers

All paths are relative to `/home/claude/CLBI_work_2026-10-24/`. "Scores file" means `outputs_v12/CLBI_v1_scores_all500.csv`.

- **409 ranked, 24 groups, 29 groups.** Count of non-missing `rank_rev`; number of distinct `score_group` among them; number of distinct `score_group` overall.
- **Boundary counts.** `brsr_reporting_boundary` in the scores file, all rows and rows with `rank_rev` present.
- **Confidence tiers (Table 3.2).** `CLBI_v1_company_data.csv`, rows for the nine core fields with `flag` "reported" or "reported_zero", cross-tabulated by `source_status`.
- **221 and 85 ranked companies on repaired values.** For ranked companies, the set of `source_status` values over their nine core fields; counted where the set is not only tier 1, and where it contains `GAPFILL_public_report`.
- **Coverage (Table 3.3).** Present: `{e}` not null; zero: `{e}_zero_flag`; held back: `{e}_query_flag` and not zero; revenue not matched: present, not zero, not held and `revenue_ok` false; usable: `{e}_per_rev` not null; scored: `{e}_score_rev` not null.
- **Scope 2 basis and Scope 3.** `scope2_basis.classification` and `metrics.scope3_tco2e.status` in `normalized_companies.json` (465 filings).
- **Implied Scope 2 factor (433; median 0.710; 286 at 0.71; 42 outside 0.5–1.0, 34 ranked).** Scope 2 (scores file) ÷ (XBRL `nonrenewable_electricity_gj` ÷ 3.6), for companies whose XBRL total energy is within 2% of the value finally used and with both terms positive (447 agree; 433 tested).
- **Offset statistics (59; 10.6%; 81.6%; 174; 41; 215).** Ranked companies: `renewable_energy_gj` ÷ `total_energy_gj` ≥ 0.5 and its median; `waste_recovered_tonnes` ÷ `waste_generated_tonnes`, median, ≥ 0.9, ≥ 1.0; `water` < `water_withdrawal_m3`.
- **Peer-group statistics (54 moves; 342; 95.3%; 361/128/11; 14).** `inputs/peer_groups_v12.csv`: `home_sector` ≠ `nse_sector`; `pg_source_url` containing "xbrl"; median of numeric `main_activity_share_pct`; `pg_confidence`; `conglomerate` = yes.
- **Rule V5 counts (7; 1; 3; 3).** Implied grid GJ > 1.5 × `total_energy_gj` (7); of which the bill rule already fires (1); of the remaining 6, those failing the credible-energy condition (3) and those held (3).
- **Rule V7 (30; 24).** `water_discharge_assumed` true; among ranked.
- **Naming caution (31; 12; 7; 10 flags; 14) and V12 (15; 14).** Ranked companies with non-empty `naming_caution`; with non-empty `business_model_flag` (from `peer_fit_notes.csv`); whose caution contains a "x above" or "x below peer median" flag, and the count of each pressure name in those flags; with `scope1_fuel_check` true (all companies: 15; ranked: 14).
- **Ties (30 in 6 groups).** Ranked companies whose `rank_rev` is shared with another company in the same group.
- **Percentile base wider than ranked set (climate 14, energy 15, water 14, pollution 9 groups).** For each pressure, groups where the count of non-missing `{e}_score_rev` exceeds the number ranked.
- **Second view of the 91 (Section 8).** Each company assigned to the first failed stage in the order: no XBRL file and incomplete; required pressure missing; revenue; open query; zero; group or sector size; incomplete group; ranked. The XBRL list is the 465 symbols in `normalized_companies.json`.
- **Nature-cost shares and medians (Section 11).** Shares: sum of `cost_ghg`, `cost_energy`, `cost_water`, `cost_waste`, `cost_air` over the 465 companies with `nature_cost_central` present, each divided by the sum of all five; for lens A the water cost is scaled by 39.6/358 and for lens C the air cost is recomputed at ₹10.2, 3.4 and 15.9 lakh per tonne of SOx, NOx and PM. Medians of `nature_cost_pct_ebitda` (and `_lens_A`, `_lens_C`) over the 370 non-null rows; medians of `nature_cost_pct_revenue` (460 rows) and of each lens total divided by revenue. Median of `share_ghg`; largest component per row of the `share_*` columns; counts above 100%. The one-convention price check and its set B comparison row are in `insights/prices/convention_check.csv`.
- **Land categories (Table 10.1).** `land_category` in `CLBI_v1_land_evidence.csv`, after `inputs/land_category_overrides.csv` is applied by `land_evidence_v12.py`.

- **Lineage (5,949 values; 0 without URL).** Rows of `CLBI_v1_company_data.csv` with `flag` “reported” or “reported_zero”, counting empty `source_url`.
- **Waste figures in V9.** `waste_generated_tonnes` and `waste_recovered_tonnes` in the scores file; Coal India’s share is its generated tonnes divided by the sum over all 500 (5,854,650,379 t).
- **Robustness.** Medians and minima of the three correlation columns of `CLBI_v1_robustness.csv` over its 24 rows (insurance has no profit correlation).

Appendix F. Review and response

An independent review of data and method (Fable, 28 October 2026; `reviews/FABLE_V12_DATA_METHOD.md`) reproduced every rank and element score of the earlier build from the gross inputs and checked the nature-cost totals. The table lists its HIGH and MEDIUM findings and our response.

Finding	What the reviewer found	Our response in this build
H1 Scope 1 contradicts fuel data	Scope 1 above what the reported fuel could emit, including three companies whose zero Scope 2 was accepted (BEL, POWERINDIA, CUPID)	We added check V12: Scope 1 above 0.12 t per GJ of non-renewable energy, outside process-emission groups and against a usable energy figure, triggers a naming caution (15 companies, 14 ranked). V12 cautions and does not hold. **Open:** whether V12 should hold the value, and whether a zero Scope 2 should be accepted only when V12 passes
H2 Price set B not on one convention	Water ₹358/kL is a PPP-converted, global-perspective IFVI value, while carbon is converted at market exchange rates; BRSR consumption can include seawater	**Resolved.** We kept set B (decision of 28 October). A recomputation with every price from one standard (IFVI and CE Delft), converted by one convention, gives the same mix by PPP or by market exchange rate (carbon 71.6–71.9%, water 15.0–15.1%), so the conclusions hold under either convention (Section 11). We now remove seawater inside reported consumption from priced water for 7 companies, read from each company's BRSR; we leave six unclear cases unadjusted and name them.
H3 Leaders that will be challenged; false cautions	Jewellers ranked among appliance makers; UPL's standalone boundary; drug makers with very low Scope 1; a delivery platform among hotels; NIC rule labelled refiners as traders	We retired the NIC-based caution. Cautions now come from reviewed notes on main activity (`peer_fit_notes.csv`, 13 companies, 12 of them ranked), V12 and the 100–300× value flags; 31 ranked companies carry one. Peer profiles name a company if it has no caution. **Open:** CESC (distribution utility among generators) and GRAVITA (lead recycler among miners) carry no reviewed note
M1 Pollution scored in real estate	Construction debris decides the top of real estate (Oberoi Realty net 0.007 t on 333,517 t generated)	We added real estate to the groups where we do not score pollution (rule V9). Real estate now ranks 9 companies on climate, energy and water
M2 Exact-20% water rule	The rule changes leaders and catches only a ratio of exactly 0.20; other round ratios are not caught and no filing has been read for "assumed"	**Kept** (decision of 28 October). We disclose its narrow reach in V7 and list it as a limitation in Section 13
M3 % of EBITDA fragile at company level	Values of several hundred per cent, partly driven by flagged inputs; thin margins	We present the EBITDA share as peer-group medians with IQR, show company values with their flags and low–high range, and flag margins below 5% (8 companies). We did not adopt "years of EBITDA" for values above 100%
M4 Lineage gaps	Seven ranked companies without a source URL on scored values; some gap-fill rows without a page; unit-converted quotes; PRIVISCL consumption still in litres	Every reported value now carries a source URL (5,949 of 5,949). **Open:** 41 gap-fill rows still lack a page; PRIVISCL's water remains held; quotes in other units remain labelled "quoted verbatim"
M5 Land categories regex-driven	Single sites, a land-acquisition claim, leases, an allotment and land banks were in the top category	We classified the 28 top-category rows by hand (Table 10.1): A 13, R 6, B 7, with 2 moved to C. We corrected BEL and BHEL to totals